

Qualitative Coding Evidence Structure

RQ	Theme	Code	Quotation
RQ1	Controlled Connectivity	Controlled remote & maintenance access	EXP-01 [Q5]: It also depends on how the software is implemented, because that is process-specific. But generally, if you have a connection toward an untrusted network, the best choice is to send only information and alarms there. There should not be direct control of anything. If control from outside is allowed, then it needs to go through a firewall, and you need to control who communicates with us and what kind of commands they can give. It also depends on which commands are allowed. Usually, you might allow a command to release a section valve, but you cannot directly command how the pumping works. It goes through our automation system before it reaches the pump units. So, yes, somebody might be able to manipulate a command and release a section valve. That is unfortunate, but it does not compromise the whole system. EXP-01 [Q12]: That is going to be tricky. It depends on the customer. You would really need to explain why you need access through the maintenance path. On some projects, we have previously delivered a dedicated remote-access device inside the cabinet, so that kind of maintenance connection has been done before. The monitoring part is more difficult. Customers may not want to allow us to have an external monitoring connection, even if we provide our own connection point, because the shipowner or system integrator takes responsibility for the overall cybersecurity of the vessel. They want to monitor and control the point where external connections leave the vessel. If there are, for example, hundreds of onboard systems and every supplier has its own external communication link, it becomes a nightmare for the system integrator. If the customer wants this kind of monitoring, then it can be done, but it will depend on their architecture, how much it is used, how much data is sent, and whether the data flow is one-way or EXP-02 [Q14]: Cybersecurity will always be a concern for people whenever there is an internet connection to somewhere. The question is how it can be done so that it is as secure as possible. But you have already covered many of these points in your thesis. There is also the business side: how can this be monetised, is it expensive, and who pays for it? What kind of data can we receive? Could data from, for example, pump units be provided or sold to the original equipment manufacturers? EXP-03 [Q3]: Yes. This is an interesting point because the normal industrial practice is that the safest way to protect the system is not to connect it to the internet at all. Realistically, though, with this framework you could potentially convince the customer to move toward a connected system if the benefits are clear. For example, the customer could get real-time monitoring, and maintenance-level access could support diagnostics and troubleshooting without giving direct operational control. That could reduce system downtime, so there is a realistic benefit that could help justify the connection. There is also a wider concern. If the customer allows external connections not only for the fire-suppression system but also for many other critical onboard systems, the ship could end up with many separate external connection points. Even if every connection is protected, having many gateways increases the overall exposure. If one of them is compromised, it could become a serious problem for the vessel. EXP-05 [Q1]: It looks concise and good. It contains the basic things we have, including connection to third-party systems. EXP-05 [Q2]: I think it is quite simplified. EXP-05 [Q4]: Yes. Temporary access should also be considered when service personnel go to the customer site. EXP-06 [Q3/Q4]: What we share with the customer should mainly be read-only information. Indications, operational data, alarms, and similar information can be shared. But when it comes to maintenance functions or making modifications, I think that access should be kept strictly under supplier control. EXP-06 [Q6]: Authentication and access control: yes. Network separation or segmentation: I am not an expert in that field, but I think yes. Restricted communication belongs to the same area and should be fine. Controlled maintenance access should also be possible. Monitoring and logging may become more challenging once the data moves outside our own system. Up to a certain point we can keep monitoring and logging within the supplier system, but in many cases the customer will be managing the data they receive. Then questions arise about how they log it, how we can use it, and how the monitoring responsibility is divided. Configuration and change management is doable, but it requires a lot of effort. The marine industry is still quite conservative, and these kinds of practices are not equally common across all systems, especially on older vessels. For protection of controller configurations, I am not sure enough to comment in detail. Controlled remote connectivity is always a challenge with moving assets such as vessels, but connectivity onboard ships has taken a significant leap during the last ten years and I think that development will continue. In the EXP-06 [Q12/Q14]: Historically, the company has sold hardware and technical devices. A solution like this requires a change in mindset because we also need to learn to sell services. The discussion with the customer should not focus on which box or device is installed. It should focus on what remote connectivity enables and what benefits it creates for both the customer and the supplier. It should be a win-win situation. Remote capabilities could be integrated into the broader service offering, for example preventive maintenance. Then the remote functionality becomes one part of the service package, helping both the supplier and the customer and potentially creating cost savings. In that model, we are selling an ongoing service rather than only transactional service visits. EXP-08 [Q6/Q7]: Yes. It is especially important that the maintenance path is unplugged or air-gapped when it is not in use. EXP-10 [Q2]: Yes, the layers make sense. The architecture starts from the field instruments, then moves to the upper levels, and finally includes the external connection possibilities. Although the system itself is complex, the diagram represents it very clearly.

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			EXP-10 [Q6/Q7]: Yes, definitely. Remote maintenance access is something we should already have in 2026, and similar capabilities are already used elsewhere. If implemented with clearly separated security layers and physical connection points that can be isolated, it would be useful for both the supplier and the customer. EXP-10 [Q12]: The concept itself is sufficient, but approvals are an important part of implementation. Customers also need the supplier system to fit into the vessel's complete cybersecurity architecture. The required evidence and documentation are defined by the relevant classification-society rules, so the requirements need to be reviewed class by class. Type approval or an equivalent recognised approval would make deployment easier and could reduce repeated project-specific work. One point I would consider is whether remote access could use a mobile network while the vessel is in port. Ships visit ports frequently, so a separate mobile connection could provide another maintenance-access option without relying on the vessel's own internet connection, provided that such an arrangement is permitted and securely implemented. EXP-11 [Q4]: Yes. I think the communication paths are clear. The way they are represented, including the different connection types, makes the communication between the areas easy to understand. EXP-11 [Q6]: Yes. Most of these measures are realistic, and several are already present in some form. We already use password protection for certain functions, controlled VPN-based access for remote connections, network separation, restricted communication, internal monitoring and logging, version and change management, and protection of controller configurations. Recovery is currently supported through local backups, such as backup storage media. Controlled remote connectivity is also realistic. Some of the exact authentication mechanisms may still need further development, but overall the measures are achievable. EXP-11 [Q7]: Yes, it is realistic. Similar remote-access arrangements have already been implemented in land-based applications using comparable controller technology, so the basic approach is feasible. EXP-12 [Q4]: For monitoring and connected services, I would prefer one-way communication out of the protected system. If an external system must provide a control input, that input should be very tightly limited. For example, an external fire-detection system may be required to request the release of a defined protected area, but it should not receive broad control authority over the fire-suppression system. For IoT, monitoring, and similar connectivity, the normal data path should be one-way. The separate remote-maintenance path is different because it needs bidirectional communication, but that connection should exist only for a defined period and should be established onboard when it is needed and agreed. That separation was well represented in the EXP-12 [Q12]: Remote connectivity changes the approval case the most. Micro-segmentation using managed or programmable network devices may also be needed and can make the system more configurable and secure. The remote-connectivity part is less mature, so it would need to be updated, explained, and assessed with the classification societies, especially where remote actions are possible. The framework gives a good starting point, but those remote functions need explicit assessment.
	Data handling & ownership		EXP-01 [Q5]: It also depends on how the software is implemented, because that is process-specific. But generally, if you have a connection toward an untrusted network, the best choice is to send only information and alarms there. There should not be direct control of anything. If control from outside is allowed, then it needs to go through a firewall, and you need to control who communicates with us and what kind of commands they can give. It also depends on which commands are allowed. Usually, you might allow a command to release a section valve, but you cannot directly command how the pumping works. It goes through our automation system before it reaches the pump units. So, yes, somebody might be able to manipulate a command and release a section valve. That is unfortunate, but it does not compromise the whole system. EXP-02 [Q12]: I do not think so, because the data is just from our products that are onboard. We already get access to that information periodically, around every six or seven months, so I do not see that as a problem. EXP-02 [Q14]: Cybersecurity will always be a concern for people whenever there is an internet connection to somewhere. The question is how it can be done so that it is as secure as possible. But you have already covered many of these points in your thesis. There is also the business side: how can this be monetised, is it expensive, and who pays for it? What kind of data can we receive? Could data from, for example, pump units be provided or sold to the original equipment manufacturers? EXP-03 [Q11]: Yes. For now, it would be a good basis for developing a system. In the longer term, as I mentioned, I think critical onboard systems may increasingly connect through a common protected ship-level server, and the fire-suppression system could then exchange selected data through that interface. EXP-05 [Q5]: It will work by itself. We can also use the manual releases. EXP-05 [Q6]: The data connection should support real-time data, with information stored locally and available for extraction when needed. EXP-06 [Q6]: Authentication and access control: yes. Network separation or segmentation: I am not an expert in that field, but I think yes. Restricted communication belongs to the same area and should be fine. Controlled maintenance access should also be possible. Monitoring and logging may become more challenging once the data moves outside our own system. Up to a certain point we can keep monitoring and logging within the supplier system, but in many cases the customer will be managing the data they receive. Then questions arise about how they log it, how we can use it, and how the monitoring responsibility is divided. Configuration and change management is doable, but it requires a lot of effort. The marine industry is still quite conservative, and these kinds of practices are not equally common across all systems, especially on older vessels. For protection of controller configurations, I am not sure enough to comment in detail. Controlled remote connectivity is always a challenge with moving assets such as vessels, but connectivity onboard ships has taken a significant leap during the last ten years and I think that development will continue. In the

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			EXP-06 [Q7]: Yes, mainly because this also requires a cultural change and new skill sets. We need to think about what capabilities we need to build in-house. It is not enough just to have the data; we also need a strategy for how to use it, people who can interpret it, and the capability to turn it into useful opportunities and service cases. These are challenges, but they are not impossible to solve. EXP-09 [Q7]: The main concern is the possibility of a security breach when access is provided. Normally we do not want to give customers broad access to the system. If access is limited mainly to monitoring, the users are clearly responsible, and passwords and permissions are properly managed, then it should be manageable. It would still create some additional responsibility for the after-sales organisation, and it would probably not be suitable for every vessel or every customer. EXP-12 [Q13]: In addition to micro-segmentation, the main open issue is remote access, especially a permanent monitoring connection. On-demand maintenance access is less concerning because a person is involved and the connection can be physically or procedurally disconnected. A permanent connection raises more questions: how strong the one-way gateway or data-diode arrangement is, how the firewall is configured, who manages the data, how access is controlled, and how credentials or VPN tokens are managed. That is probably the weakest or, at least, the least-defined part at the moment. EXP-12 [Q14]: It depends on what the service is trying to achieve. If the data can be processed locally, and the result is needed locally, then I would process and store it locally where possible. Beyond that, I do not think there is one fixed answer yet. The future service model will determine what data is generated, where it is stored, what the customer wants to receive, and whether the customer accesses it through an API or a fleet-level monitoring system. My expectation is that customers will increasingly want interfaces to their own monitoring environments rather than manage all raw data onboard themselves. The supplier may process some data and provide selected results or analytics, while the customer consumes those results through its own systems. The exact
	External access protection		EXP-01 [Q1]: Yeah. In my opinion, maybe a bit too much. There are too many layers and too many zones. Of course, it is quite a heavy topic because we talk about layers and we talk about zones. I think the layers are more like, in a software way, where you do the stuff, while the zones are more physical. Part of that also goes to the system integrator, which is the shipowner or shipbuilder. If you first think about the zones, I do not think we need a separate fire-suppression zone; we could just have the safety-function zone. The maintenance and engineering zone is also a bit much. I would almost lose it or place those functions between the supervisory and safety areas. The fire-suppression equipment itself should already be in a user-access-controlled place. It is not for everybody. You should not allow everybody to come there, supervise it, or use it. In the same way, the operational panels should also be in restricted areas. And communication between a third party and our system should be quite isolated from everything else. EXP-01 [Q2]: Yes. In the best scenario, we do not have that at all. If we do have it, then we need a firewall and we need to define very clearly how communication works across that interface. EXP-01 [Q4]: You should also talk about the physical locations of the equipment. Later, you talk about user profiles, roles, access, passwords, and things like that. But for fire suppression and the panels, they should always be available without first needing to log in. At the same time, they need to be located somewhere where people cannot just manipulate them. That is why we advise the integrator that they need to be located in user-access-controlled areas, such as the bridge or the ECR. That way, some of the password-management burden can be reduced. You should mention that the equipment, whether it is an electrical cabinet, HMI panel, pump unit, or something similar, should be in an access-controlled area so that not everybody can access it without EXP-01 [Q5]: It also depends on how the software is implemented, because that is process-specific. But generally, if you have a connection toward an untrusted network, the best choice is to send only information and alarms there. There should not be direct control of anything. If control from outside is allowed, then it needs to go through a firewall, and you need to control who communicates with us and what kind of commands they can give. It also depends on which commands are allowed. Usually, you might allow a command to release a section valve, but you cannot directly command how the pumping works. It goes through our automation system before it reaches the pump units. So, yes, somebody might be able to manipulate a command and release a section valve. That is unfortunate, but it does not compromise the whole system. EXP-01 [Q6]: Those are more on the software side, for example SCADA and PLC systems. The cybersecurity requirements talk about user profiles and passwords, but for the fire-suppression panel, it should always be available so that you can immediately release or operate the system. You should not need to log in first. That is one of the difficult points when applying the standards. At the moment, we do not really have full role-based access. We mainly have one service access level for doing updates or similar work, and we are even considering whether that could be avoided. Passwords also create additional requirements: password length, failed-login handling, logging of failed attempts, and so on. All of that adds requirements to the software. For normal operation, release is done physically onboard the vessel and does not require a password because the equipment is already in a user-access-controlled area. It is not sitting on an open deck where anybody can access it. So there is physical security as well. EXP-01 [Q8/Q10]: Yes. We have already had our cybersecurity arrangements audited by a classification-related third party, and we passed. There was also discussion about Modbus TCP/IP communication because release signals were coming from a third party into our system. The important point was what could actually be changed from outside. You cannot directly change how the pump unit operates or its firefighting capacity. You may be able to affect whether a section is opened or not, so you can do some harm, but not the kind of harm that compromises the complete system. That is how we do it.

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			EXP-01 [Q10]: A lot relies on the equipment being installed in user-access-controlled areas, and that is something we cannot fully control. We can only give guidance. Once the system is delivered, we cannot control what somebody does with the electrical cabinet or where exactly it is installed. We do not monitor whether somebody physically opens the electrical cabinet, but we do monitor the network. If somebody unplugs a cable, for example, we get an alarm. So cable breaks or attempts to manipulate the network EXP-01 [Q11]: I think it is quite a good pipeline. We have already done many of these things, so not all of it is new to us, but it brings together many things that need to be considered. Especially the cybersecurity part is a good summary of what is listed in the standards, such as audit logging and monitoring. If you have user roles or device access, and people log in or log out, those events should be recorded in the logs. Password requirements and related security events also need to be handled. EXP-01 [Q13]: Maybe simplify the zones and layers a little bit. EXP-01 [Q14]: Try to highlight the physical user access as well. The architecture is currently more focused on the software side, but it is quite simple to add the requirement that access points to the system should be located in user-access-controlled areas. There is also the practical issue of physical ports. The standard says that unused open ports should be disabled. In practice, if a panel is installed inside a console or another user-access-controlled location, physically reaching and manipulating that connection may already require a significant amount of effort. So the physical-access EXP-02 [Q3/Q4]: I am wondering whether it is necessary to have so many separate access levels. EXP-02 [Q4]: Yeah, for sure. I do not know exactly what the roles should be, but it would be good to consider different customer profiles or different vessel profiles. They can have different teams and different access needs. EXP-02 [Q3]: One thing I was thinking about was the security level. Could it be shown as going from low to high? That could simplify the diagram a lot for me. It would make it easier to understand if the untrusted-network zone is clearly the lower-security area and the fire-suppression zone is clearly the high-security area. I would perhaps also add who has access to each of these areas, for example which onboard personnel have access to each security level or zone. EXP-02 [Q10]: To be honest, that was my main concern when I first thought about this: access, because the software is connected to the internet. But if there is only one-way data access, so it is only receiving the data in one direction, that mitigates my concern. EXP-02 [Q14]: Cybersecurity will always be a concern for people whenever there is an internet connection to somewhere. The question is how it can be done so that it is as secure as possible. But you have already covered many of these points in your thesis. There is also the business side: how can this be monetised, is it expensive, and who pays for it? What kind of data can we receive? Could data from, for example, pump units be provided or sold to the original equipment manufacturers? EXP-03 [Q4]: As a stand-alone supplier system, the framework is very good. But in the future, I think ships may combine the externally connected critical systems through one protected ship-level connection point or server. That server could collect the necessary information from systems such as fire suppression, machinery, and energy monitoring, and then external applications or suppliers would obtain selected data through that protected ship-level interface. In my view, that could be stronger than every individual vendor maintaining its own external connection, because the vessel or system integrator can protect and control one EXP-03 [Q7]: Yes. If the network is segmented into multiple subnets, it may create difficulties for external or third-party systems that need to access specific data. It is no longer as simple as pointing directly to the IP address of an end device. The third-party system may need to reach a particular controller or data point through a different subnet, and that can make configuration and data access EXP-03 [Q11]: Yes. For now, it would be a good basis for developing a system. In the longer term, as I mentioned, I think critical onboard systems may increasingly connect through a common protected ship-level server, and the fire-suppression system could then exchange selected data through that interface. EXP-04 [Q10]: Yes. If it is sufficiently protected, I think it could. EXP-04 [Q12]: I think they would receive it quite positively. EXP-04 [Q12]: The biggest problem is that there are no references yet. It is difficult to sell something that has not been used before. Cybersecurity is very important, so larger companies could definitely be interested, especially if the price and the financial case are EXP-05 [Q2]: I think it is quite simplified. EXP-05 [Q4]: Yes. Temporary access should also be considered when service personnel go to the customer site. EXP-05 [Q4/Q6]: We only want to access our own system. HMI panels can be viewed locally, while PLC software should only be accessible with password protection. EXP-05 [Q13/Q14]: There is always a risk when connecting to a network. Networks should therefore be isolated even inside the ship, for example by using a Modbus RTU serial connection where appropriate. EXP-06 [Q3/Q4]: What we share with the customer should mainly be read-only information. Indications, operational data, alarms, and similar information can be shared. But when it comes to maintenance functions or making modifications, I think that access should be kept strictly under supplier control.

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			EXP-06 [Q6]: Authentication and access control: yes. Network separation or segmentation: I am not an expert in that field, but I think yes. Restricted communication belongs to the same area and should be fine. Controlled maintenance access should also be possible. Monitoring and logging may become more challenging once the data moves outside our own system. Up to a certain point we can keep monitoring and logging within the supplier system, but in many cases the customer will be managing the data they receive. Then questions arise about how they log it, how we can use it, and how the monitoring responsibility is divided. Configuration and change management is doable, but it requires a lot of effort. The marine industry is still quite conservative, and these kinds of practices are not equally common across all systems, especially on older vessels. For protection of controller configurations, I am not sure enough to comment in detail. Controlled remote connectivity is always a challenge with moving assets such as vessels, but connectivity onboard ships has taken a significant leap during the last ten years and I think that development will continue. In the EXP-07 [Q4]: The main thing is separation or segmentation of the control network from other systems. When integration between the fire-suppression system and external systems is required, we need to consider the security of those connections and prevent unauthorised access through those communication paths. Network segmentation is probably the most important point. Remote connections and maintenance pathways also have to be designed and controlled very carefully. EXP-07 [Q6/Q7]: It can be, but quite often customers require the fire-suppression system to remain fully isolated from the internet. That is an important practical limitation and depends very much on the vessel or site. EXP-07 [Q6]: Yes, I think most of these measures can be practically implemented. In some areas, however, we depend heavily on the equipment manufacturers. Certain required features may be limited by the capabilities of the devices being used. For example, with configuration and change management, we may not be able to implement device-level tracking of configuration changes if the equipment does not support it. In that case, we have to manage the change-control process on our own side rather than relying on EXP-07 [Q14]: There are many possible ways to implement the architecture, and the details are highly customer-specific and manufacturer-specific. Another supplier could make different design decisions and still follow the same basic idea. The common principle should remain the same: first isolate the control network from the outside world as much as possible, and then add any necessary external connections in a controlled and secure way without compromising that basic isolation principle. EXP-08 [Q6/Q7]: Yes. It is especially important that the maintenance path is unplugged or air-gapped when it is not in use. EXP-08 [Q14]: From my experience, the core system that makes the fire-suppression system functional without any outside connections should be clearly separated from all outside connections. Cybersecurity measures should first be designed properly around that core system, and only then should outside connections be added while keeping those principles intact. In practice, an important preventive measure is to keep the core equipment inside access-controlled spaces. The system should also detect any physical tampering. If someone gains access to a control cabinet, unplugs something, or connects a new device, that should be detected immediately and should generate an alarm so that someone can intervene. For the core equipment itself, user logins may not always be the most important protection. If physical access to the equipment is already tightly controlled and any interaction or tampering is immediately detected, that can be more important than adding further hardening that could complicate essential EXP-09 [Q7]: The main concern is the possibility of a security breach when access is provided. Normally we do not want to give customers broad access to the system. If access is limited mainly to monitoring, the users are clearly responsible, and passwords and permissions are properly managed, then it should be manageable. It would still create some additional responsibility for the after-sales organisation, and it would probably not be suitable for every vessel or every customer. EXP-09 [Q12]: For non-technical and commercial people, I would want a very simple diagram showing what the current system looks like and what is being added in the future. It should be clear that this is not a completely new fire-suppression system; it is additional cybersecurity and connectivity around the existing system. The simpler that comparison is, the easier it will be to explain internally and to customers. When presenting to a customer, the relevant electrical or engineering people can then review the technical details. A cybersecurity and connected-service offering could provide commercial value because not every competitor offers the same capability or certification. It could be sold as an additional option for new projects and potentially as an upgrade for the existing installed base. Even if the additional hardware cost per vessel is relatively modest, applying the upgrade across a large EXP-10 [Q3]: Yes, I think so. If the access between the different levels can be properly protected, including password protection where it is technically possible, I think the system can be secured. The practical question is whether all of these separate access controls can actually be implemented between the different system levels. EXP-10 [Q4]: For some customers this will still be a new concept. It therefore needs to be explained very clearly: what we are going to do, how we are going to do it, and why it is needed. The implementation will also require contribution from the customer because the supplier system becomes part of the vessel's overall cybersecurity architecture. EXP-11 [Q6]: Yes. Most of these measures are realistic, and several are already present in some form. We already use password protection for certain functions, controlled VPN-based access for remote connections, network separation, restricted communication, internal monitoring and logging, version and change management, and protection of controller configurations. Recovery is currently supported through local backups, such as backup storage media. Controlled remote connectivity is also realistic. Some of the exact authentication mechanisms may still need further development, but overall the measures are achievable.

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			EXP-12 [Q3]: Fire suppression is part of the safety function in my opinion. The system has a fairly specific purpose: fire suppression. I would therefore consider combining the separate safety-function and fire-suppression areas into one fire-suppression functionality zone. If they remain separate, the contents of each zone need to be described more clearly. The supervisory area could be intended for authorised supplier personnel, while maintenance or engineering access from a third party should be lower-trust and more EXP-12 [Q6]: In principle, yes, but the controls need to be applied at the right level. For a connected monitoring or cloud service, stronger user management and cybersecurity controls are clearly needed. At the device level, however, implementing local user management on PLC-based equipment may add too much complexity and may not even be practical with the current device capabilities. There is also a trade-off. If local user management requires a connection to supplier servers or to the vessel IT environment, that new connection can itself introduce additional risk. It could also turn each project into a broader IT integration project. For simple equipment-level control of pumps, valves, and local controllers, it may therefore be better to keep the control system simpler and use other protective measures rather than force every whole-vessel IT requirement down to the device level. EXP-12 [Q7]: They can create unnecessary complexity if they are applied without considering the actual function of the system. The standards cover the whole vessel and many different types of information and access-control needs. A fire-suppression system is primarily controlling pumps, valves, and equipment-level controllers, so not every whole-vessel control is necessarily needed locally. The architecture therefore needs a balance between cybersecurity requirements, operational simplicity, and preservation of the safety function. The situation changes when the system uses a connected monitoring application or cloud service. Once public-internet or cloud connectivity is introduced, the cybersecurity requirements become more demanding. EXP-12 [Q12]: Remote connectivity changes the approval case the most. Micro-segmentation using managed or programmable network devices may also be needed and can make the system more configurable and secure. The remote-connectivity part is less mature, so it would need to be updated, explained, and assessed with the classification societies, especially where remote actions are possible. The framework gives a good starting point, but those remote functions need explicit assessment. EXP-12 [Q13]: In addition to micro-segmentation, the main open issue is remote access, especially a permanent monitoring connection. On-demand maintenance access is less concerning because a person is involved and the connection can be physically or procedurally disconnected. A permanent connection raises more questions: how strong the one-way gateway or data-diode arrangement is, how the firewall is configured, who manages the data, how access is controlled, and how credentials or VPN tokens are managed. That is probably the weakest or, at least, the least-defined part at the moment. EXP-12 [Q14]: It depends on what the service is trying to achieve. If the data can be processed locally, and the result is needed locally, then I would process and store it locally where possible. Beyond that, I do not think there is one fixed answer yet. The future service model will determine what data is generated, where it is stored, what the customer wants to receive, and whether the customer accesses it through an API or a fleet-level monitoring system. My expectation is that customers will increasingly want interfaces to their own monitoring environments rather than manage all raw data onboard themselves. The supplier may process some data and provide selected results or analytics, while the customer consumes those results through its own systems. The exact
	One-way / read-only monitoring		EXP-01 [Q2]: Yes. It could be combined into a monitoring and supervisory area, or something similar. EXP-01 [Q5]: It also depends on how the software is implemented, because that is process-specific. But generally, if you have a connection toward an untrusted network, the best choice is to send only information and alarms there. There should not be direct control of anything. If control from outside is allowed, then it needs to go through a firewall, and you need to control who communicates with us and what kind of commands they can give. It also depends on which commands are allowed. Usually, you might allow a command to release a section valve, but you cannot directly command how the pumping works. It goes through our automation system before it reaches the pump units. So, yes, somebody might be able to manipulate a command and release a section valve. That is unfortunate, but it does not compromise the whole system. EXP-01 [Q12]: That is going to be tricky. It depends on the customer. You would really need to explain why you need access through the maintenance path. On some projects, we have previously delivered a dedicated remote-access device inside the cabinet, so that kind of maintenance connection has been done before. The monitoring part is more difficult. Customers may not want to allow us to have an external monitoring connection, even if we provide our own connection point, because the shipowner or system integrator takes responsibility for the overall cybersecurity of the vessel. They want to monitor and control the point where external connections leave the vessel. If there are, for example, hundreds of onboard systems and every supplier has its own external communication link, it becomes a nightmare for the system integrator. If the customer wants this kind of monitoring, then it can be done, but it will depend on their architecture, how much it is used, how much data is sent, and whether the data flow is one-way or EXP-02 [Q10]: To be honest, that was my main concern when I first thought about this: access, because the software is connected to the internet. But if there is only one-way data access, so it is only receiving the data in one direction, that mitigates my concern. EXP-02 [Q12]: I do not think so, because the data is just from our products that are onboard. We already get access to that information periodically, around every six or seven months, so I do not see that as a problem.

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			EXP-03 [Q3]: Yes. This is an interesting point because the normal industrial practice is that the safest way to protect the system is not to connect it to the internet at all. Realistically, though, with this framework you could potentially convince the customer to move toward a connected system if the benefits are clear. For example, the customer could get real-time monitoring, and maintenance-level access could support diagnostics and troubleshooting without giving direct operational control. That could reduce system downtime, so there is a realistic benefit that could help justify the connection. There is also a wider concern. If the customer allows external connections not only for the fire-suppression system but also for many other critical onboard systems, the ship could end up with many separate external connection points. Even if every connection is protected, having many gateways increases the overall exposure. If one of them is compromised, it could become a serious problem for the vessel. EXP-04 [Q14]: One thing I would ask about is the monitoring application: who would use it, and who would be responsible for it? EXP-05 [Q4/Q6]: We only want to access our own system. HMI panels can be viewed locally, while PLC software should only be accessible with password protection. EXP-06 [Q3/Q4]: What we share with the customer should mainly be read-only information. Indications, operational data, alarms, and similar information can be shared. But when it comes to maintenance functions or making modifications, I think that access should be kept strictly under supplier control. EXP-09 [Q7]: The main concern is the possibility of a security breach when access is provided. Normally we do not want to give customers broad access to the system. If access is limited mainly to monitoring, the users are clearly responsible, and passwords and permissions are properly managed, then it should be manageable. It would still create some additional responsibility for the after-sales organisation, and it would probably not be suitable for every vessel or every customer. EXP-11 [Q1]: Yes. The structure looks very clear to me. It represents the current type of system quite well, but now it is organised into different layers, with the monitoring and maintenance functions added on top. EXP-11 [Q7]: Yes, it is realistic. Similar remote-access arrangements have already been implemented in land-based applications using comparable controller technology, so the basic approach is feasible. EXP-12 [Q4]: For monitoring and connected services, I would prefer one-way communication out of the protected system. If an external system must provide a control input, that input should be very tightly limited. For example, an external fire-detection system may be required to request the release of a defined protected area, but it should not receive broad control authority over the fire-suppression system. For IoT, monitoring, and similar connectivity, the normal data path should be one-way. The separate remote-maintenance path is different because it needs bidirectional communication, but that connection should exist only for a defined period and should be established onboard when it is needed and agreed. That separation was well represented in the EXP-12 [Q13]: In addition to micro-segmentation, the main open issue is remote access, especially a permanent monitoring connection. On-demand maintenance access is less concerning because a person is involved and the connection can be physically or procedurally disconnected. A permanent connection raises more questions: how strong the one-way gateway or data-diode arrangement is, how the firewall is configured, who manages the data, how access is controlled, and how credentials or VPN tokens are managed. That is probably the weakest or, at least, the least-defined part at the moment.
	Marine System Integration	Customer / vessel integration	EXP-01 [Q2]: Yeah. The standards are a little tricky because, if I remember correctly, they do not really say that you need to have these exact zones. You need to define them yourself. When we talk with a classification society, they may have some examples, but they are only examples. They are not saying that you must have exactly these zones. They recommend separating things into EXP-02 [Q3/Q4]: I am wondering whether it is necessary to have so many separate access levels. EXP-02 [Q4]: Yeah, for sure. I do not know exactly what the roles should be, but it would be good to consider different customer profiles or different vessel profiles. They can have different teams and different access needs. EXP-03 [Q2]: Not at the moment. The layers look quite good. I do not have any further comment on that. We have also discussed that this is a general framework and deliberately does not reproduce a confidential company-specific architecture, so for that purpose I would say it is very solid. In an implementation-specific version, some of the naming or allocation of functions and zones EXP-03 [Q3]: Yes. This is an interesting point because the normal industrial practice is that the safest way to protect the system is not to connect it to the internet at all. Realistically, though, with this framework you could potentially convince the customer to move toward a connected system if the benefits are clear. For example, the customer could get real-time monitoring, and maintenance-level access could support diagnostics and troubleshooting without giving direct operational control. That could reduce system downtime, so there is a realistic benefit that could help justify the connection. There is also a wider concern. If the customer allows external connections not only for the fire-suppression system but also for many other critical onboard systems, the ship could end up with many separate external connection points. Even if every connection is protected, having many gateways increases the overall exposure. If one of them is compromised, it could become a serious problem for the vessel. EXP-03 [Q4]: As a stand-alone supplier system, the framework is very good. But in the future, I think ships may combine the externally connected critical systems through one protected ship-level connection point or server. That server could collect the necessary information from systems such as fire suppression, machinery, and energy monitoring, and then external applications or suppliers would obtain selected data through that protected ship-level interface. In my view, that could be stronger than every individual vendor maintaining its own external connection, because the vessel or system integrator can protect and control one

RQ	Theme	Code	Quotation
			EXP-03 [Q11]: Yes. For now, it would be a good basis for developing a system. In the longer term, as I mentioned, I think critical onboard systems may increasingly connect through a common protected ship-level server, and the fire-suppression system could then exchange selected data through that interface. EXP-04 [Q13]: The benefits need to be made clearer. It should be explained in which situations this would be the best decision for the customer. EXP-04 [Q14]: One thing I would ask about is the monitoring application: who would use it, and who would be responsible for it? EXP-05 [Q1]: It looks concise and good. It contains the basic things we have, including connection to third-party systems. EXP-05 [Q2]: I think it is quite simplified. EXP-05 [Q4]: Yes. Temporary access should also be considered when service personnel go to the customer site. EXP-06 [Q2]: The main concern for me is the interface. Especially in retrofit projects, there can be many different applications, system suppliers, and existing arrangements onboard. The interface toward customer systems is therefore important. I am not saying it is missing from the framework, but when the concept moves toward implementation, I think that interface may be the EXP-06 [Q4]: We need to be careful and pay a lot of attention to how we present the solution. I also think it is important to standardise the offering as far as possible. If we have many different tailor-made solutions and offers, it becomes very laborious for the customer to evaluate them and understand whether they meet the customer's own cybersecurity policies and requirements. EXP-06 [Q12/Q14]: Historically, the company has sold hardware and technical devices. A solution like this requires a change in mindset because we also need to learn to sell services. The discussion with the customer should not focus on which box or device is installed. It should focus on what remote connectivity enables and what benefits it creates for both the customer and the supplier. It should be a win-win situation. Remote capabilities could be integrated into the broader service offering, for example preventive maintenance. Then the remote functionality becomes one part of the service package, helping both the supplier and the customer and potentially creating cost savings. In that model, we are selling an ongoing service rather than only transactional service visits. EXP-06 [Q12]: The customer approach varies a lot, and it depends on who the point of contact is. In my experience, these cybersecurity and remote-connectivity discussions often need to involve different people than the technical contacts we normally work with. The people responsible for cybersecurity need to be involved so that they can discuss the requirements with people who speak the same technical language. At some point, the people working on the technical side also need to be involved. A concrete demonstration like the one you showed is very useful because then people can see what the concept actually means. If we only show diagrams and layout drawings, the solution is not very tangible, it is harder to understand, and progress can be slow. EXP-06 [Q13]: We need to follow the industry closely and understand how the standards and requirements develop. The offering should remain fit for purpose and be as standardised as possible. Customer maturity is also important. There are many different kinds of customers, and they are not all equally ready for this type of solution. Some may not yet have the capabilities or the right people to work with this kind of project. So successful implementation does not depend only on the supplier; it also requires effort EXP-07 [Q6/Q7]: It can be, but quite often customers require the fire-suppression system to remain fully isolated from the internet. That is an important practical limitation and depends very much on the vessel or site. EXP-08 [Q4]: They are kind of realistic, but I do not know whether anyone would find it feasible to implement the framework exactly as shown. It would require vessel-specific design and adaptation. Systems can be implemented differently, so the same boundaries would not necessarily apply in exactly the same way to every system or vessel. EXP-08 [Q11]: Yes. Referring to the previous answer, it is at least a good starting point. These points should be considered first, and then the project should examine whether anything more is required for the specific system. EXP-09 [Q4]: We first need to understand the customer's needs, what kind of system they already have, and whether this additional functionality is needed. It would likely be offered as an optional add-on rather than as part of every standard delivery. The interface and integration would then be adapted to the customer's requirements. EXP-09 [Q11]: Yes. It represents the existing system with additional cybersecurity and connectivity functions. If cybersecurity compliance and certification requirements are incorporated, I think an industrial implementation would likely be very close to this framework, even if the final design is not exactly identical. EXP-09 [Q12]: For non-technical and commercial people, I would want a very simple diagram showing what the current system looks like and what is being added in the future. It should be clear that this is not a completely new fire-suppression system; it is additional cybersecurity and connectivity around the existing system. The simpler that comparison is, the easier it will be to explain internally and to customers. When presenting to a customer, the relevant electrical or engineering people can then review the technical details. A cybersecurity and connected-service offering could provide commercial value because not every competitor offers the same capability or certification. It could be sold as an additional option for new projects and potentially as an upgrade for the existing installed base. Even if the additional hardware cost per vessel is relatively modest, applying the upgrade across a large EXP-10 [Q4]: For some customers this will still be a new concept. It therefore needs to be explained very clearly: what we are going to do, how we are going to do it, and why it is needed. The implementation will also require contribution from the customer because the supplier system becomes part of the vessel's overall cybersecurity architecture. EXP-10 [Q5]: Yes, from the supplier-system perspective it would. The core system should continue to operate without the external connection. However, the complete solution also depends on the customer's implementation and contribution on the vessel side.

RQ	Theme	Code	Quotation
			EXP-10 [Q11]: Yes. The concepts are very similar to the systems used in practice: there are field instruments, controllers, networks, and the higher-level functions. The framework could therefore be useful for future industrial projects. EXP-10 [Q12]: The concept itself is sufficient, but approvals are an important part of implementation. Customers also need the supplier system to fit into the vessel's complete cybersecurity architecture. The required evidence and documentation are defined by the relevant classification-society rules, so the requirements need to be reviewed class by class. Type approval or an equivalent recognised approval would make deployment easier and could reduce repeated project-specific work. One point I would consider is whether remote access could use a mobile network while the vessel is in port. Ships visit ports frequently, so a separate mobile connection could provide another maintenance-access option without relying on the vessel's own internet connection, provided that such an arrangement is permitted and securely implemented. EXP-10 [Q14]: The concept looks very good. The main next step is to go through the cybersecurity rules of the relevant classification societies and determine what approvals are required. A reusable type approval or similarly recognised approval could make later vessel implementations easier and reduce repeated approval work. EXP-12 [Q12]: Customers will want confidence that the system can obtain the required IACS UR E27 compliance evidence or certification and that the documentation is complete. For a fire-suppression system, many customers are not yet asking for extensive IoT or remote-monitoring functionality; that is still relatively new. The demand is likely to increase as customers become more interested in mobile or fleet-level monitoring. EXP-12 [Q14]: I think the main points have been covered. The important thing is that this field is continuously evolving. Traditional industrial control systems are moving from relay- and PLC-centred technology toward more software- and IT-oriented architectures. I expect the industrial-control and IT worlds to continue converging, so the cybersecurity architecture will also need to evolve with EXP-12 [Q14]: It depends on what the service is trying to achieve. If the data can be processed locally, and the result is needed locally, then I would process and store it locally where possible. Beyond that, I do not think there is one fixed answer yet. The future service model will determine what data is generated, where it is stored, what the customer wants to receive, and whether the customer accesses it through an API or a fleet-level monitoring system. My expectation is that customers will increasingly want interfaces to their own monitoring environments rather than manage all raw data onboard themselves. The supplier may process some data and provide selected results or analytics, while the customer consumes those results through its own systems. The exact
	Practical, approval & business constraints		EXP-01 [Q1]: Yeah. In my opinion, maybe a bit too much. There are too many layers and too many zones. Of course, it is quite a heavy topic because we talk about layers and we talk about zones. I think the layers are more like, in a software way, where you do the stuff, while the zones are more physical. Part of that also goes to the system integrator, which is the shipowner or shipbuilder. If you first think about the zones, I do not think we need a separate fire-suppression zone; we could just have the safety-function zone. The maintenance and engineering zone is also a bit much. I would almost lose it or place those functions between the supervisory and safety areas. The fire-suppression equipment itself should already be in a user-access-controlled place. It is not for everybody. You should not allow everybody to come there, supervise it, or use it. In the same way, the operational panels should also be in restricted areas. And communication between a third party and our system should be quite isolated from everything else. EXP-01 [Q2]: Yeah. The standards are a little tricky because, if I remember correctly, they do not really say that you need to have these exact zones. You need to define them yourself. When we talk with a classification society, they may have some examples, but they are only examples. They are not saying that you must have exactly these zones. They recommend separating things into EXP-01 [Q8/Q10]: Yes. We have already had our cybersecurity arrangements audited by a classification-related third party, and we passed. There was also discussion about Modbus TCP/IP communication because release signals were coming from a third party into our system. The important point was what could actually be changed from outside. You cannot directly change how the pump unit operates or its firefighting capacity. You may be able to affect whether a section is opened or not, so you can do some harm, but not the kind of harm that compromises the complete system. That is how we do it. EXP-02 [Q14]: Cybersecurity will always be a concern for people whenever there is an internet connection to somewhere. The question is how it can be done so that it is as secure as possible. But you have already covered many of these points in your thesis. There is also the business side: how can this be monetised, is it expensive, and who pays for it? What kind of data can we receive? Could data from, for example, pump units be provided or sold to the original equipment manufacturers? EXP-03 [Q2]: Not at the moment. The layers look quite good. I do not have any further comment on that. We have also discussed that this is a general framework and deliberately does not reproduce a confidential company-specific architecture, so for that purpose I would say it is very solid. In an implementation-specific version, some of the naming or allocation of functions and zones EXP-03 [Q6]: In principle, yes, but the practicality varies a lot from customer to customer and from vessel to vessel. Technically, it is feasible. The implementation question is how you convince the customer to spend additional money on it and what benefit the system brings to them. There is an economic aspect as well as the technical one. EXP-04 [Q10]: Yes. If it is sufficiently protected, I think it could. EXP-04 [Q12]: I think they would receive it quite positively. EXP-04 [Q12]: It can be done, but it needs to be explained clearly, separated into understandable parts, and made simpler for the customer. Not everyone who needs to understand it will have a technical background.

RQ	Theme	Code	Quotation
			EXP-04 [Q12]: The biggest problem is that there are no references yet. It is difficult to sell something that has not been used before. Cybersecurity is very important, so larger companies could definitely be interested, especially if the price and the financial case are EXP-04 [Q13]: The benefits need to be made clearer. It should be explained in which situations this would be the best decision for the customer EXP-05 [Q1]: It looks concise and good. It contains the basic things we have, including connection to third-party systems. EXP-05 [Q11]: It could be implemented for most vessels, except military vessels. EXP-05 [Q12]: For yachts, privacy may also be a EXP-06 [Q2]: The main concern for me is the interface. Especially in retrofit projects, there can be many different applications, system suppliers, and existing arrangements onboard. The interface toward customer systems is therefore important. I am not saying it is missing from the framework, but when the concept moves toward implementation, I think that interface may be the EXP-06 [Q4]: We need to be careful and pay a lot of attention to how we present the solution. I also think it is important to standardise the offering as far as possible. If we have many different tailor-made solutions and offers, it becomes very laborious for the customer to evaluate them and understand whether they meet the customer's own cybersecurity policies and requirements. EXP-06 [Q5]: It is hard to say in general because we have systems installed on many different vessels, and in some cases compromises or limitations will be necessary. The age of the vessel, the available investment, where the vessel operates, and what kind of connectivity systems are already installed onboard all affect what can realistically be implemented. I would see newbuild and retrofit projects as two different journeys. On new vessels, these kinds of cybersecurity and connectivity requirements can be considered already during the ship design. On older vessels, the variation between installations is much larger. EXP-06 [Q6]: Authentication and access control: yes. Network separation or segmentation: I am not an expert in that field, but I think yes. Restricted communication belongs to the same area and should be fine. Controlled maintenance access should also be possible. Monitoring and logging may become more challenging once the data moves outside our own system. Up to a certain point we can keep monitoring and logging within the supplier system, but in many cases the customer will be managing the data they receive. Then questions arise about how they log it, how we can use it, and how the monitoring responsibility is divided. Configuration and change management is doable, but it requires a lot of effort. The marine industry is still quite conservative, and these kinds of practices are not equally common across all systems, especially on older vessels. For protection of controller configurations, I am not sure enough to comment in detail. Controlled remote connectivity is always a challenge with moving assets such as vessels, but connectivity onboard ships has taken a significant leap during the last ten years and I think that development will continue. In the EXP-06 [Q7]: Yes, mainly because this also requires a cultural change and new skill sets. We need to think about what capabilities we need to build in-house. It is not enough just to have the data; we also need a strategy for how to use it, people who can interpret it, and the capability to turn it into useful opportunities and service cases. These are challenges, but they are not impossible to solve. EXP-06 [Q12/Q14]: Historically, the company has sold hardware and technical devices. A solution like this requires a change in mindset because we also need to learn to sell services. The discussion with the customer should not focus on which box or device is installed. It should focus on what remote connectivity enables and what benefits it creates for both the customer and the supplier. It should be a win-win situation. Remote capabilities could be integrated into the broader service offering, for example preventive maintenance. Then the remote functionality becomes one part of the service package, helping both the supplier and the customer and potentially creating cost savings. In that model, we are selling an ongoing service rather than only transactional service visits. EXP-06 [Q13]: We need to follow the industry closely and understand how the standards and requirements develop. The offering should remain fit for purpose and be as standardised as possible. Customer maturity is also important. There are many different kinds of customers, and they are not all equally ready for this type of solution. Some may not yet have the capabilities or the right people to work with this kind of project. So successful implementation does not depend only on the supplier; it also requires effort EXP-07 [Q2]: Not much. There is one point that may only be about how the schematic is represented. The connection to the data server and the maintenance gateway could physically be one device while remaining logically separated. That may simply be a matter of presentation and clarity. Otherwise, I do not see unnecessary complexity. I think the framework is detailed enough that implementation work could be started from it. EXP-07 [Q6]: Yes, I think most of these measures can be practically implemented. In some areas, however, we depend heavily on the equipment manufacturers. Certain required features may be limited by the capabilities of the devices being used. For example, with configuration and change management, we may not be able to implement device-level tracking of configuration changes if the equipment does not support it. In that case, we have to manage the change-control process on our own side rather than relying on EXP-07 [Q11]: Yes. I think it would be very useful and could be used as a baseline or starting point for an industrial project. EXP-07 [Q12]: If I were using this as a basis for a specific system, I would define more clearly what each zone and layer means in that particular implementation. The general concept is good, but the definitions need to be made more specific for the actual system and EXP-08 [Q4]: They are kind of realistic, but I do not know whether anyone would find it feasible to implement the framework exactly as shown. It would require vessel-specific design and adaptation. Systems can be implemented differently, so the same boundaries would not necessarily apply in exactly the same way to every system or vessel. EXP-08 [Q6]: Yes. In general, if a particular measure is applicable to the system, it should also be implementable.

RQ	Theme	Code	Quotation
			EXP-08 [Q11]: Yes. Referring to the previous answer, it is at least a good starting point. These points should be considered first, and then the project should examine whether anything more is required for the specific system. EXP-09 [Q2]: I do not think it looks too complicated for the people who would actually handle the system. They normally already understand this type of system, so it should be relatively easy for them to learn. Tailoring can be useful when possible, but from our side standardisation would also be preferable. EXP-09 [Q4]: We first need to understand the customer's needs, what kind of system they already have, and whether this additional functionality is needed. It would likely be offered as an optional add-on rather than as part of every standard delivery. The interface and integration would then be adapted to the customer's requirements. EXP-09 [Q11]: Yes. It represents the existing system with additional cybersecurity and connectivity functions. If cybersecurity compliance and certification requirements are incorporated, I think an industrial implementation would likely be very close to this framework, even if the final design is not exactly identical. EXP-09 [Q12]: For non-technical and commercial people, I would want a very simple diagram showing what the current system looks like and what is being added in the future. It should be clear that this is not a completely new fire-suppression system; it is additional cybersecurity and connectivity around the existing system. The simpler that comparison is, the easier it will be to explain internally and to customers. When presenting to a customer, the relevant electrical or engineering people can then review the technical details. A cybersecurity and connected-service offering could provide commercial value because not every competitor offers the same capability or certification. It could be sold as an additional option for new projects and potentially as an upgrade for the existing installed base. Even if the additional hardware cost per vessel is relatively modest, applying the upgrade across a large EXP-10 [Q3]: Yes, I think so. If the access between the different levels can be properly protected, including password protection where it is technically possible, I think the system can be secured. The practical question is whether all of these separate access controls can actually be implemented between the different system levels. EXP-10 [Q4]: For some customers this will still be a new concept. It therefore needs to be explained very clearly: what we are going to do, how we are going to do it, and why it is needed. The implementation will also require contribution from the customer because the supplier system becomes part of the vessel's overall cybersecurity architecture. EXP-10 [Q12]: The concept itself is sufficient, but approvals are an important part of implementation. Customers also need the supplier system to fit into the vessel's complete cybersecurity architecture. The required evidence and documentation are defined by the relevant classification-society rules, so the requirements need to be reviewed class by class. Type approval or an equivalent recognised approval would make deployment easier and could reduce repeated project-specific work. One point I would consider is whether remote access could use a mobile network while the vessel is in port. Ships visit ports frequently, so a separate mobile connection could provide another maintenance-access option without relying on the vessel's own internet connection, provided that such an arrangement is permitted and securely implemented. EXP-10 [Q14]: The concept looks very good. The main next step is to go through the cybersecurity rules of the relevant classification societies and determine what approvals are required. A reusable type approval or similarly recognised approval could make later vessel implementations easier and reduce repeated approval work. EXP-11 [Q7]: Yes, it is realistic. Similar remote-access arrangements have already been implemented in land-based applications using comparable controller technology, so the basic approach is feasible. EXP-12 [Q1]: Yes. I think it describes very well what we have learned and what has been used in recent systems. It contains what I consider relevant at the moment. EXP-12 [Q7]: They can create unnecessary complexity if they are applied without considering the actual function of the system. The standards cover the whole vessel and many different types of information and access-control needs. A fire-suppression system is primarily controlling pumps, valves, and equipment-level controllers, so not every whole-vessel control is necessarily needed locally. The architecture therefore needs a balance between cybersecurity requirements, operational simplicity, and preservation of the safety function. The situation changes when the system uses a connected monitoring application or cloud service. Once public-internet or cloud connectivity is introduced, the cybersecurity requirements become more demanding. EXP-12 [Q12]: Customers will want confidence that the system can obtain the required IACS UR E27 compliance evidence or certification and that the documentation is complete. For a fire-suppression system, many customers are not yet asking for extensive IoT or remote-monitoring functionality; that is still relatively new. The demand is likely to increase as customers become more interested in mobile or fleet-level monitoring. EXP-12 [Q12]: Remote connectivity changes the approval case the most. Micro-segmentation using managed or programmable network devices may also be needed and can make the system more configurable and secure. The remote-connectivity part is less mature, so it would need to be updated, explained, and assessed with the classification societies, especially where remote actions are possible. The framework gives a good starting point, but those remote functions need explicit assessment. EXP-12 [Q14]: I think the main points have been covered. The important thing is that this field is continuously evolving. Traditional industrial control systems are moving from relay- and PLC-centred technology toward more software- and IT-oriented architectures. I expect the industrial-control and IT worlds to continue converging, so the cybersecurity architecture will also need to evolve with

RQ	Theme	Code	Quotation
RQ2	Cyber Threats & Attack Paths	Data, signal or configuration manipulation	EXP-01 [Q8/Q10]: Yes. We have already had our cybersecurity arrangements audited by a classification-related third party, and we passed. There was also discussion about Modbus TCP/IP communication because release signals were coming from a third party into our system. The important point was what could actually be changed from outside. You cannot directly change how the pump unit operates or its firefighting capacity. You may be able to affect whether a section is opened or not, so you can do some harm, but not the kind of harm that compromises the complete system. That is how we do it. EXP-01 [Q10]: A lot relies on the equipment being installed in user-access-controlled areas, and that is something we cannot fully control. We can only give guidance. Once the system is delivered, we cannot control what somebody does with the electrical cabinet or where exactly it is installed. We do not monitor whether somebody physically opens the electrical cabinet, but we do monitor the network. If somebody unplugs a cable, for example, we get an alarm. So cable breaks or attempts to manipulate the network EXP-03 [Q3]: Yes. This is an interesting point because the normal industrial practice is that the safest way to protect the system is not to connect it to the internet at all. Realistically, though, with this framework you could potentially convince the customer to move toward a connected system if the benefits are clear. For example, the customer could get real-time monitoring, and maintenance-level access could support diagnostics and troubleshooting without giving direct operational control. That could reduce system downtime, so there is a realistic benefit that could help justify the connection. There is also a wider concern. If the customer allows external connections not only for the fire-suppression system but also for many other critical onboard systems, the ship could end up with many separate external connection points. Even if every connection is protected, having many gateways increases the overall exposure. If one of them is compromised, it could become a serious problem for the vessel. EXP-09 [Q9]: I cannot comment in detail on the technical threat analysis, but ship systems in general can be compromised. One practical concern is weak password protection. I have seen situations where a customer was able to gain access to a password-protected controller even though they were not intended to have that level of access. That suggests that controller credentials and access protection need to be treated very carefully. EXP-12 [Q6]: In principle, yes, but the controls need to be applied at the right level. For a connected monitoring or cloud service, stronger user management and cybersecurity controls are clearly needed. At the device level, however, implementing local user management on PLC-based equipment may add too much complexity and may not even be practical with the current device capabilities. There is also a trade-off. If local user management requires a connection to supplier servers or to the vessel IT environment, that new connection can itself introduce additional risk. It could also turn each project into a broader IT integration project. For simple equipment-level control of pumps, valves, and local controllers, it may therefore be better to keep the control system simpler and use other protective measures rather than force every whole-vessel IT requirement down to the device level.
		DoS / communication disruption	EXP-01 [Q10]: A lot relies on the equipment being installed in user-access-controlled areas, and that is something we cannot fully control. We can only give guidance. Once the system is delivered, we cannot control what somebody does with the electrical cabinet or where exactly it is installed. We do not monitor whether somebody physically opens the electrical cabinet, but we do monitor the network. If somebody unplugs a cable, for example, we get an alarm. So cable breaks or attempts to manipulate the network EXP-03 [Q4]: As a stand-alone supplier system, the framework is very good. But in the future, I think ships may combine the externally connected critical systems through one protected ship-level connection point or server. That server could collect the necessary information from systems such as fire suppression, machinery, and energy monitoring, and then external applications or suppliers would obtain selected data through that protected ship-level interface. In my view, that could be stronger than every individual vendor maintaining its own external connection, because the vessel or system integrator can protect and control one
		Malware / third-party compromise	EXP-05 [Q4/Q9]: Someone could compromise a service computer and, when that computer is taken onsite for service, malware could be introduced into the system. EXP-08 [Q2]: Pretty much. At least the core system that generates alarms or performs actuation could be considered trusted. Everything outside the fire-suppression system supplier's own delivery could then be considered untrusted.
		Physical intrusion & attack propagation	EXP-01 [Q4]: You should also talk about the physical locations of the equipment. Later, you talk about user profiles, roles, access, passwords, and things like that. But for fire suppression and the panels, they should always be available without first needing to log in. At the same time, they need to be located somewhere where people cannot just manipulate them. That is why we advise the integrator that they need to be located in user-access-controlled areas, such as the bridge or the ECR. That way, some of the password-management burden can be reduced. You should mention that the equipment, whether it is an electrical cabinet, HMI panel, pump unit, or something similar, should be in an access-controlled area so that not everybody can access it without EXP-01 [Q6]: Those are more on the software side, for example SCADA and PLC systems. The cybersecurity requirements talk about user profiles and passwords, but for the fire-suppression panel, it should always be available so that you can immediately release or operate the system. You should not need to log in first. That is one of the difficult points when applying the standards. At the moment, we do not really have full role-based access. We mainly have one service access level for doing updates or similar work, and we are even considering whether that could be avoided. Passwords also create additional requirements: password length, failed-login handling, logging of failed attempts, and so on. All of that adds requirements to the software. For normal operation, release is done physically onboard the vessel and does not require a password because the equipment is already in a user-access-controlled area. It is not sitting on an open deck where anybody can access it. So there is physical security as well.

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			EXP-01 [Q10]: A lot relies on the equipment being installed in user-access-controlled areas, and that is something we cannot fully control. We can only give guidance. Once the system is delivered, we cannot control what somebody does with the electrical cabinet or where exactly it is installed. We do not monitor whether somebody physically opens the electrical cabinet, but we do monitor the network. If somebody unplugs a cable, for example, we get an alarm. So cable breaks or attempts to manipulate the network EXP-01 [Q13]: Maybe simplify the zones and layers a little bit. EXP-01 [Q14]: Try to highlight the physical user access as well. The architecture is currently more focused on the software side, but it is quite simple to add the requirement that access points to the system should be located in user-access-controlled areas. There is also the practical issue of physical ports. The standard says that unused open ports should be disabled. In practice, if a panel is installed inside a console or another user-access-controlled location, physically reaching and manipulating that connection may already require a significant amount of effort. So the physical-access EXP-07 [Q8]: Yes. I think these are realistic cases that could actually happen. EXP-07 [Q9]: I think the main priority should be the security of access from the untrusted zone. That is the biggest concern. On a passenger vessel, the equipment is normally installed in spaces that passengers cannot access. A crew member could potentially gain physical access and connect something directly to the network, but that becomes more of a physical-security issue and is probably less likely than an attack through an external or
		Unauthorized / remote access	EXP-01 [Q1]: Yeah. In my opinion, maybe a bit too much. There are too many layers and too many zones. Of course, it is quite a heavy topic because we talk about layers and we talk about zones. I think the layers are more like, in a software way, where you do the stuff, while the zones are more physical. Part of that also goes to the system integrator, which is the shipowner or shipbuilder. If you first think about the zones, I do not think we need a separate fire-suppression zone; we could just have the safety-function zone. The maintenance and engineering zone is also a bit much. I would almost lose it or place those functions between the supervisory and safety areas. The fire-suppression equipment itself should already be in a user-access-controlled place. It is not for everybody. You should not allow everybody to come there, supervise it, or use it. In the same way, the operational panels should also be in restricted areas. And communication between a third party and our system should be quite isolated from everything else. EXP-01 [Q5]: It also depends on how the software is implemented, because that is process-specific. But generally, if you have a connection toward an untrusted network, the best choice is to send only information and alarms there. There should not be direct control of anything. If control from outside is allowed, then it needs to go through a firewall, and you need to control who communicates with us and what kind of commands they can give. It also depends on which commands are allowed. Usually, you might allow a command to release a section valve, but you cannot directly command how the pumping works. It goes through our automation system before it reaches the pump units. So, yes, somebody might be able to manipulate a command and release a section valve. That is unfortunate, but it does not compromise the whole system. EXP-01 [Q6]: Those are more on the software side, for example SCADA and PLC systems. The cybersecurity requirements talk about user profiles and passwords, but for the fire-suppression panel, it should always be available so that you can immediately release or operate the system. You should not need to log in first. That is one of the difficult points when applying the standards. At the moment, we do not really have full role-based access. We mainly have one service access level for doing updates or similar work, and we are even considering whether that could be avoided. Passwords also create additional requirements: password length, failed-login handling, logging of failed attempts, and so on. All of that adds requirements to the software. For normal operation, release is done physically onboard the vessel and does not require a password because the equipment is already in a user-access-controlled area. It is not sitting on an open deck where anybody can access it. So there is physical security as well. EXP-01 [Q8/Q10]: Yes. We have already had our cybersecurity arrangements audited by a classification-related third party, and we passed. There was also discussion about Modbus TCP/IP communication because release signals were coming from a third party into our system. The important point was what could actually be changed from outside. You cannot directly change how the pump unit operates or its firefighting capacity. You may be able to affect whether a section is opened or not, so you can do some harm, but not the kind of harm that compromises the complete system. That is how we do it. EXP-01 [Q9]: One thing is the internal network itself. During the audit, they also monitored whether we had unused services or ports open on the internal network. That should be considered as well. EXP-01 [Q12]: That is going to be tricky. It depends on the customer. You would really need to explain why you need access through the maintenance path. On some projects, we have previously delivered a dedicated remote-access device inside the cabinet, so that kind of maintenance connection has been done before. The monitoring part is more difficult. Customers may not want to allow us to have an external monitoring connection, even if we provide our own connection point, because the shipowner or system integrator takes responsibility for the overall cybersecurity of the vessel. They want to monitor and control the point where external connections leave the vessel. If there are, for example, hundreds of onboard systems and every supplier has its own external communication link, it becomes a nightmare for the system integrator. If the customer wants this kind of monitoring, then it can be done, but it will depend on their architecture, how much data is used, how much data is sent, and whether the data flow is one-way or EXP-02 [Q3]: One thing I was thinking about was the security level. Could it be shown as going from low to high? That could simplify the diagram a lot for me. It would make it easier to understand if the untrusted-network zone is clearly the lower-security area and the fire-suppression zone is clearly the high-security area. I would perhaps also add who has access to each of these areas, for example which onboard personnel have access to each security level or zone.

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			EXP-03 [Q3]: Yes. This is an interesting point because the normal industrial practice is that the safest way to protect the system is not to connect it to the internet at all. Realistically, though, with this framework you could potentially convince the customer to move toward a connected system if the benefits are clear. For example, the customer could get real-time monitoring, and maintenance-level access could support diagnostics and troubleshooting without giving direct operational control. That could reduce system downtime, so there is a realistic benefit that could help justify the connection. There is also a wider concern. If the customer allows external connections not only for the fire-suppression system but also for many other critical onboard systems, the ship could end up with many separate external connection points. Even if every connection is protected, having many gateways increases the overall exposure. If one of them is compromised, it could become a serious problem for the vessel. EXP-05 [Q2]: I think it is quite simplified. EXP-05 [Q4]: Yes. Temporary access should also be considered when service personnel go to the customer site. EXP-07 [Q3]: Yes, I think it is mostly realistic. One difficulty is that the Safety Function Zone and the physical fire-suppression functions cannot always be fully separated in a real implementation. In this particular case, if you gain access to some safety-control functions, that may also provide access toward the fire-suppression functions. So that boundary can be difficult to realise exactly as shown, but otherwise the separation is realistic. EXP-07 [Q8]: Yes. I think these are realistic cases that could actually happen. EXP-07 [Q9]: I think the main priority should be the security of access from the untrusted zone. That is the biggest concern. On a passenger vessel, the equipment is normally installed in spaces that passengers cannot access. A crew member could potentially gain physical access and connect something directly to the network, but that becomes more of a physical-security issue and is probably less likely than an attack through an external or EXP-08 [Q2]: Pretty much. At least the core system that generates alarms or performs actuation could be considered trusted. Everything outside the fire-suppression system supplier's own delivery could then be considered untrusted. EXP-08 [Q8]: Yes, they look realistic. The assessment may not be complete, because it may not be possible to imagine every type of threat in advance. EXP-09 [Q7]: The main concern is the possibility of a security breach when access is provided. Normally we do not want to give customers broad access to the system. If access is limited mainly to monitoring, the users are clearly responsible, and passwords and permissions are properly managed, then it should be manageable. It would still create some additional responsibility for the after-sales organisation, and it would probably not be suitable for every vessel or every customer. EXP-09 [Q9]: I cannot comment in detail on the technical threat analysis, but ship systems in general can be compromised. One practical concern is weak password protection. I have seen situations where a customer was able to gain access to a password-protected controller even though they were not intended to have that level of access. That suggests that controller credentials and access protection need to be treated very carefully. EXP-12 [Q6]: In principle, yes, but the controls need to be applied at the right level. For a connected monitoring or cloud service, stronger user management and cybersecurity controls are clearly needed. At the device level, however, implementing local user management on PLC-based equipment may add too much complexity and may not even be practical with the current device capabilities. There is also a trade-off. If local user management requires a connection to supplier servers or to the vessel IT environment, that new connection can itself introduce additional risk. It could also turn each project into a broader IT integration project. For simple equipment-level control of pumps, valves, and local controllers, it may therefore be better to keep the control system simpler and use other protective measures rather than force every whole-vessel IT requirement down to the device level.
RQ3	Recovery	Recovery & restoration	EXP-01 [Q5]: Yes. There should always be a backup if the automation fails. EXP-01 [Q5]: I think it is quite robust. The main risk is the untrusted-network connection, but otherwise, no. EXP-01 [Q6/Q7]: Those are possible. Some of them already come from the components and controller platform itself, such as traffic filtering. On the recovery side, we also have redundancy, and we provide software backups so the software can be restored if EXP-05 [Q5]: It will work by itself. We can also use the manual releases. EXP-05 [Q6]: The data connection should support real-time data, with information stored locally and available for extraction when needed. EXP-07 [Q5]: Yes. That is also the current default approach. EXP-07 [Q5]: No, I cannot see an obvious single point of failure in the EXP-11 [Q6]: Yes. Most of these measures are realistic, and several are already present in some form. We already use password protection for certain functions, controlled VPN-based access for remote connections, network separation, restricted communication, internal monitoring and logging, version and change management, and protection of controller configurations. Recovery is currently supported through local backups, such as backup storage media. Controlled remote connectivity is also realistic. Some of the exact authentication mechanisms may still need further development, but overall the measures are achievable. EXP-12 [Q9]: No. I think you have covered the main scenarios well. EXP-12 [Q9]: Micro-segmentation should receive more attention. Denial-of-service is also something that should be considered. Recovery and restoration are especially important, and not only for cybersecurity. The same capability is needed for hardware or equipment failures so that people onboard can restore the system and replace failed equipment.

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	Resilience & Containment	Fault & alarm visibility	EXP-01 [Q10]: A lot relies on the equipment being installed in user-access-controlled areas, and that is something we cannot fully control. We can only give guidance. Once the system is delivered, we cannot control what somebody does with the electrical cabinet or where exactly it is installed. We do not monitor whether somebody physically opens the electrical cabinet, but we do monitor the network. If somebody unplugs a cable, for example, we get an alarm. So cable breaks or attempts to manipulate the network EXP-01 [Q11]: I think it is quite a good pipeline. We have already done many of these things, so not all of it is new to us, but it brings together many things that need to be considered. Especially the cybersecurity part is a good summary of what is listed in the standards, such as audit logging and monitoring. If you have user roles or device access, and people log in or log out, those events should be recorded in the logs. Password requirements and related security events also need to be handled. EXP-08 [Q14]: From my experience, the core system that makes the fire-suppression system functional without any outside connections should be clearly separated from all outside connections. Cybersecurity measures should first be designed properly around that core system, and only then should outside connections be added while keeping those principles intact. In practice, an important preventive measure is to keep the core equipment inside access-controlled spaces. The system should also detect any physical tampering. If someone gains access to a control cabinet, unplugs something, or connects a new device, that should be detected immediately and should generate an alarm so that someone can intervene. For the core equipment itself, user logins may not always be the most important protection. If physical access to the equipment is already tightly controlled and any interaction or tampering is immediately detected, that can be more important than adding further hardening that could complicate essential EXP-12 [Q5]: The micro-segmentation needs to be defined more clearly in the architecture. Another important point is alarm and degraded-state awareness. If a cyber incident or component loss occurs but the crew does not receive an alarm, the system has effectively failed in another way because people do not know that manual action may be required. Redundant controllers therefore need to be configured so that failures are detected and an alarm clearly indicates that the system is no longer fully functional. EXP-12 [Q12]: First, I would define the micro-segmentation more clearly. Second, I would link the architecture directly to the applicable requirements. If we have an IT or control-system topology and a cybersecurity design, I want to be able to point from each relevant IEC 62443 requirement to the architectural feature that addresses it. The same traceability is important for IACS UR E27 and classification-society approval. If the architecture is taken into use, the next step is to build the approval documentation from those
		Independent / degraded / manual operation	EXP-01 [Q1]: Yeah. In my opinion, maybe a bit too much. There are too many layers and too many zones. Of course, it is quite a heavy topic because we talk about layers and we talk about zones. I think the layers are more like, in a software way, where you do the stuff, while the zones are more physical. Part of that also goes to the system integrator, which is the shipowner or shipbuilder. If you first think about the zones, I do not think we need a separate fire-suppression zone; we could just have the safety-function zone. The maintenance and engineering zone is also a bit much. I would almost lose it or place those functions between the supervisory and safety areas. The fire-suppression equipment itself should already be in a user-access-controlled place. It is not for everybody. You should not allow everybody to come there, supervise it, or use it. In the same way, the operational panels should also be in restricted areas. And communication between a third party and our system should be quite isolated from everything else. EXP-01 [Q5]: Exactly. It is a false activation. The pump is still available, and if there is a real alarm somewhere else, the system can still operate. Even if our automation system goes down, the local panels can still be used to manually open the section valve, and the pump unit can still be operated manually. The fire-suppression capability is therefore not compromised. EXP-01 [Q5]: It also depends on how the software is implemented, because that is process-specific. But generally, if you have a connection toward an untrusted network, the best choice is to send only information and alarms there. There should not be direct control of anything. If control from outside is allowed, then it needs to go through a firewall, and you need to control who communicates with us and what kind of commands they can give. It also depends on which commands are allowed. Usually, you might allow a command to release a section valve, but you cannot directly command how the pumping works. It goes through our automation system before it reaches the pump units. So, yes, somebody might be able to manipulate a command and release a section valve. That is unfortunate, but it does not compromise the whole system. EXP-01 [Q5]: Yes. There should always be a backup if the automation fails. EXP-01 [Q5]: I think it is quite robust. The main risk is the untrusted-network connection, but otherwise, no. EXP-01 [Q6]: Those are more on the software side, for example SCADA and PLC systems. The cybersecurity requirements talk about user profiles and passwords, but for the fire-suppression panel, it should always be available so that you can immediately release or operate the system. You should not need to log in first. That is one of the difficult points when applying the standards. At the moment, we do not really have full role-based access. We mainly have one service access level for doing updates or similar work, and we are even considering whether that could be avoided. Passwords also create additional requirements: password length, failed-login handling, logging of failed attempts, and so on. All of that adds requirements to the software. For normal operation, release is done physically onboard the vessel and does not require a password because the equipment is already in a user-access-controlled area. It is not sitting on an open deck where anybody can access it. So there is physical security as well. EXP-03 [Q5]: Yes. If you move all the way down through the zones and even the final safety-control level is compromised, that is already a very severe situation. Even then, the system can still operate in manual mode. There may be no automatic release, but the vessel would still retain manual fire-suppression capability. A person can detect the fire, use the manual release, and operate the

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			EXP-05 [Q5]: It will work by itself. We can also use the manual releases. EXP-05 [Q6]: The data connection should support real-time data, with information stored locally and available for extraction when needed. EXP-07 [Q3]: Yes, I think it is mostly realistic. One difficulty is that the Safety Function Zone and the physical fire-suppression functions cannot always be fully separated in a real implementation. In this particular case, if you gain access to some safety-control functions, that may also provide access toward the fire-suppression functions. So that boundary can be difficult to realise exactly as shown, but otherwise the separation is realistic. EXP-07 [Q5]: With the way the zones are defined, as long as we do not lose functionality in the Safety Function Zone and the fire-suppression functions themselves, the system can remain fully functional even if the other zones are compromised. EXP-08 [Q5]: Looking at the schematics and the principles, there appears to be enough redundancy for the safety functions. The need for redundancy is well described, so I do not see an obvious single point of failure at the conceptual level. EXP-10 [Q5]: Yes, from the supplier-system perspective it would. The core system should continue to operate without the external connection. However, the complete solution also depends on the customer's implementation and contribution on the vessel side. EXP-12 [Q5]: I would like the upper-level control and externally connected functions to be separated from the pump-unit and lower-level control functions. I would call that micro-segmentation. If the upper level, which communicates outside the system and is therefore more exposed, is lost or compromised, the lower segments should continue to operate. The individual pump units should still work. Protected areas can also be released manually, and the pump units should start automatically when required but should EXP-12 [Q5]: Yes. The system should work without external communication. That is also part of the cybersecurity testing logic. We would not normally disconnect the upper level deliberately, but a firewall may isolate communication if it detects abnormal traffic. The rest of the system must then continue to work. This is why the fire-suppression system should be independently capable and should have its own local release panel rather than depending only on a third-party system.
		Redundancy & failover	EXP-01 [Q5]: Exactly. It is a false activation. The pump is still available, and if there is a real alarm somewhere else, the system can still operate. Even if our automation system goes down, the local panels can still be used to manually open the section valve, and the pump unit can still be operated manually. The fire-suppression capability is therefore not compromised. EXP-01 [Q6/Q7]: Those are possible. Some of them already come from the components and controller platform itself, such as traffic filtering. On the recovery side, we also have redundancy, and we provide software backups so the software can be restored if EXP-03 [Q2]: First, I think you should focus more on segmentation of the entire network. Second, you should identify some of the most common risks and then run representative scenarios against the framework to see how it reacts and how it protects the EXP-03 [Q5]: Yes, but I would describe that as redundancy. Redundant controllers are not necessarily used in every system. They are more relevant for larger vessels, while smaller systems may have a single PLC and rely on manual release if that controller fails. In larger redundant systems, if one PLC fails, the system can switch to the other available PLC. A ring-network topology also provides an alternative communication path, so the switchover can still work if one communication connection is interrupted. That redundancy supports continued operation after a controller or network-path failure. EXP-03 [Q5]: Yes. If you move all the way down through the zones and even the final safety-control level is compromised, that is already a very severe situation. Even then, the system can still operate in manual mode. There may be no automatic release, but the vessel would still retain manual fire-suppression capability. A person can detect the fire, use the manual release, and operate the EXP-07 [Q5]: Yes, that is realistic. Redundancy of this kind is already an established requirement in relevant applications and has been used in systems for decades. EXP-08 [Q2]: I do not think so. There may even be more in the framework than what would actually be needed in a specific implementation. EXP-08 [Q5]: Looking at the schematics and the principles, there appears to be enough redundancy for the safety functions. The need for redundancy is well described, so I do not see an obvious single point of failure at the conceptual level. EXP-12 [Q5]: The micro-segmentation needs to be defined more clearly in the architecture. Another important point is alarm and degraded-state awareness. If a cyber incident or component loss occurs but the crew does not receive an alarm, the system has effectively failed in another way because people do not know that manual action may be required. Redundant controllers therefore need to be configured so that failures are detected and an alarm clearly indicates that the system is no longer fully functional. EXP-12 [Q5]: Yes. The system should work without external communication. That is also part of the cybersecurity testing logic. We would not normally disconnect the upper level deliberately, but a firewall may isolate communication if it detects abnormal traffic. The rest of the system must then continue to work. This is why the fire-suppression system should be independently capable and should have its own local release panel rather than depending only on a third-party system.
		Segmentation & safety separation	EXP-01 [Q2]: I would lose the maintenance and engineering zone and the separate fire-suppression zone. I would also avoid an untrusted-network zone if possible. EXP-01 [Q2]: Yes. It could be combined into a monitoring and supervisory area, or something similar.

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			EXP-01 [Q5]: It also depends on how the software is implemented, because that is process-specific. But generally, if you have a connection toward an untrusted network, the best choice is to send only information and alarms there. There should not be direct control of anything. If control from outside is allowed, then it needs to go through a firewall, and you need to control who communicates with us and what kind of commands they can give. It also depends on which commands are allowed. Usually, you might allow a command to release a section valve, but you cannot directly command how the pumping works. It goes through our automation system before it reaches the pump units. So, yes, somebody might be able to manipulate a command and release a section valve. That is unfortunate, but it does not compromise the whole system. EXP-02 [Q3]: One thing I was thinking about was the security level. Could it be shown as going from low to high? That could simplify the diagram a lot for me. It would make it easier to understand if the untrusted-network zone is clearly the lower-security area and the fire-suppression zone is clearly the high-security area. I would perhaps also add who has access to each of these areas, for example which onboard personnel have access to each security level or zone. EXP-03 [Q2]: First, I think you should focus more on segmentation of the entire network. Second, you should identify some of the most common risks and then run representative scenarios against the framework to see how it reacts and how it protects the EXP-03 [Q5]: Yes, but I would describe that as redundancy. Redundant controllers are not necessarily used in every system. They are more relevant for larger vessels, while smaller systems may have a single PLC and rely on manual release if that controller fails. In larger redundant systems, if one PLC fails, the system can switch to the other available PLC. A ring-network topology also provides an alternative communication path, so the switchover can still work if one communication connection is interrupted. That redundancy supports continued operation after a controller or network-path failure. EXP-03 [Q7]: Yes. If the network is segmented into multiple subnets, it may create difficulties for external or third-party systems that need to access specific data. It is no longer as simple as pointing directly to the IP address of an end device. The third-party system may need to reach a particular controller or data point through a different subnet, and that can make configuration and data access EXP-03 [Q9/Q10]: That comes back to the point we already discussed. You should identify the most common risks and then apply them to the framework to see how the architecture reacts. I think that scenario-based demonstration is not presented strongly enough yet. At the moment, though, the framework looks quite good and appears to protect the system against many common EXP-03 [Q12]: Network segmentation would be one of the first things to clarify. You should also present specific risk-assessment scenarios and show how the framework responds to them. Customers may not care about every technical detail of how the architecture works. They need to understand what benefit it gives them. For example, you could show a realistic attack or failure scenario and demonstrate how the framework protects the system or reduces downtime. They will also care about the economic side: how much it costs, how quickly it can be implemented, and whether it is worth adding to a system that already works well. EXP-03 [Q13]: Network segmentation needs more attention. You also need to identify specific risks and show how those risks are mitigated by the framework. EXP-05 [Q13/Q14]: There is always a risk when connecting to a network. Networks should therefore be isolated even inside the ship, for example by using a Modbus RTU serial connection where appropriate. EXP-07 [Q2]: I could combine the data-server and maintenance-gateway functions physically, as I mentioned, while keeping them logically separated. In this particular type of fire-protection system, the Communication Network Layer, Safety Function Layer, and Supervisory Monitoring Layer also have quite a lot of overlap and may be partly combined in practice. But as a conceptual way of presenting the architecture, I think the current separation is better for understanding it. EXP-07 [Q2]: Not much. There is one point that may only be about how the schematic is represented. The connection to the data server and the maintenance gateway could physically be one device while remaining logically separated. That may simply be a matter of presentation and clarity. Otherwise, I do not see unnecessary complexity. I think the framework is detailed enough that implementation work could be started from it. EXP-07 [Q4]: The main thing is separation or segmentation of the control network from other systems. When integration between the fire-suppression system and external systems is required, we need to consider the security of those connections and prevent unauthorised access through those communication paths. Network segmentation is probably the most important point. Remote connections and maintenance pathways also have to be designed and controlled very carefully. EXP-07 [Q5]: Direct physical access to the safety functions could still compromise them. Apart from that, at least at the conceptual level, it does not appear easy to compromise the essential functions if the proposed separation is maintained. EXP-07 [Q5]: With the way the zones are defined, as long as we do not lose functionality in the Safety Function Zone and the fire-suppression functions themselves, the system can remain fully functional even if the other zones are compromised. EXP-07 [Q5]: Yes, that is realistic. Redundancy of this kind is already an established requirement in relevant applications and has been used in systems for decades. EXP-07 [Q6/Q7]: It can be, but quite often customers require the fire-suppression system to remain fully isolated from the internet. That is an important practical limitation and depends very much on the vessel or site.

RQ	Theme	Code	Quotation
			EXP-07 [Q13]: The architecture itself is quite good. The main question is whether, in a specific implementation, we can actually separate all of the zones and layers as cleanly as the framework shows. For example, in our type of system, access to the Supervisory Monitoring Layer may also provide a path toward the Communication Network Layer. It is not necessarily a complete compromise, but the layers are not always independent in practice. That implementation-level overlap needs to be considered. EXP-07 [Q14]: There are many possible ways to implement the architecture, and the details are highly customer-specific and manufacturer-specific. Another supplier could make different design decisions and still follow the same basic idea. The common principle should remain the same: first isolate the control network from the outside world as much as possible, and then add any necessary external connections in a controlled and secure way without compromising that basic isolation principle. EXP-08 [Q2]: It could be simplified into trusted and untrusted areas. That would be the simplest view. EXP-08 [Q5]: Yes, if the essential safety functions are kept sufficiently and clearly separated from anything that could be compromised from an outside or untrusted network. An external compromise must not be able to compromise the core safety EXP-08 [Q14]: From my experience, the core system that makes the fire-suppression system functional without any outside connections should be clearly separated from all outside connections. Cybersecurity measures should first be designed properly around that core system, and only then should outside connections be added while keeping those principles intact. In practice, an important preventive measure is to keep the core equipment inside access-controlled spaces. The system should also detect any physical tampering. If someone gains access to a control cabinet, unplugs something, or connects a new device, that should be detected immediately and should generate an alarm so that someone can intervene. For the core equipment itself, user logins may not always be the most important protection. If physical access to the equipment is already tightly controlled and any interaction or tampering is immediately detected, that can be more important than adding further hardening that could complicate essential EXP-10 [Q6/Q7]: Yes, definitely. Remote maintenance access is something we should already have in 2026, and similar capabilities are already used elsewhere. If implemented with clearly separated security layers and physical connection points that can be isolated, it would be useful for both the supplier and the customer. EXP-10 [Q10]: Yes. In theory, the concept appears sufficient because there are clearly defined security layers and physical connection points that can be isolated. EXP-11 [Q1]: Yes. The structure looks very clear to me. It represents the current type of system quite well, but now it is organised into different layers, with the monitoring and maintenance functions added on top. EXP-11 [Q3]: At least in theory, I think so. The existing functions are already organised in a similar way, and the cyber-resilience and connectivity functions could be added around them. EXP-11 [Q6]: Yes. Most of these measures are realistic, and several are already present in some form. We already use password protection for certain functions, controlled VPN-based access for remote connections, network separation, restricted communication, internal monitoring and logging, version and change management, and protection of controller configurations. Recovery is currently supported through local backups, such as backup storage media. Controlled remote connectivity is also realistic. Some of the exact authentication mechanisms may still need further development, but overall the measures are achievable. EXP-12 [Q3]: Fire suppression is part of the safety function in my opinion. The system has a fairly specific purpose: fire suppression. I would therefore consider combining the separate safety-function and fire-suppression areas into one fire-suppression functionality zone. If they remain separate, the contents of each zone need to be described more clearly. The supervisory area could be intended for authorised supplier personnel, while maintenance or engineering access from a third party should be lower-trust and more EXP-12 [Q3]: Yes. The main schematic describes it well. I would still simplify some of the other layer or zone definitions where the separation does not provide a clear benefit. EXP-12 [Q5]: I would like the upper-level control and externally connected functions to be separated from the pump-unit and lower-level control functions. I would call that micro-segmentation. If the upper level, which communicates outside the system and is therefore more exposed, is lost or compromised, the lower segments should continue to operate. The individual pump units should still work. Protected areas can also be released manually, and the pump units should start automatically when required but should EXP-12 [Q5]: The micro-segmentation needs to be defined more clearly in the architecture. Another important point is alarm and degraded-state awareness. If a cyber incident or component loss occurs but the crew does not receive an alarm, the system has effectively failed in another way because people do not know that manual action may be required. Redundant controllers therefore need to be configured so that failures are detected and an alarm clearly indicates that the system is no longer fully functional. EXP-12 [Q7]: They can create unnecessary complexity if they are applied without considering the actual function of the system. The standards cover the whole vessel and many different types of information and access-control needs. A fire-suppression system is primarily controlling pumps, valves, and equipment-level controllers, so not every whole-vessel control is necessarily needed locally. The architecture therefore needs a balance between cybersecurity requirements, operational simplicity, and preservation of the safety function. The situation changes when the system uses a connected monitoring application or cloud service. Once public-internet or cloud connectivity is introduced, the cybersecurity requirements become more demanding.

RQ	Theme	Code	Quotation
			EXP-12 [Q9]: No. I think you have covered the main scenarios well. EXP-12 [Q9]: Micro-segmentation should receive more attention. Denial-of-service is also something that should be considered. Recovery and restoration are especially important, and not only for cybersecurity. The same capability is needed for hardware or equipment failures so that people onboard can restore the system and replace failed equipment. EXP-12 [Q12]: First, I would define the micro-segmentation more clearly. Second, I would link the architecture directly to the applicable requirements. If we have an IT or control-system topology and a cybersecurity design, I want to be able to point from each relevant IEC 62443 requirement to the architectural feature that addresses it. The same traceability is important for IACS UR E27 and classification-society approval. If the architecture is taken into use, the next step is to build the approval documentation from those EXP-12 [Q12]: Remote connectivity changes the approval case the most. Micro-segmentation using managed or programmable network devices may also be needed and can make the system more configurable and secure. The remote-connectivity part is less mature, so it would need to be updated, explained, and assessed with the classification societies, especially where remote actions are possible. The framework gives a good starting point, but those remote functions need explicit assessment.
VAL	Expert Validation	Framework supported	EXP-01 [Q2]: I would lose the maintenance and engineering zone and the separate fire-suppression zone. I would also avoid an untrusted-network zone if possible. EXP-01 [Q6]: Those are more on the software side, for example SCADA and PLC systems. The cybersecurity requirements talk about user profiles and passwords, but for the fire-suppression panel, it should always be available so that you can immediately release or operate the system. You should not need to log in first. That is one of the difficult points when applying the standards. At the moment, we do not really have full role-based access. We mainly have one service access level for doing updates or similar work, and we are even considering whether that could be avoided. Passwords also create additional requirements: password length, failed-login handling, logging of failed attempts, and so on. All of that adds requirements to the software. For normal operation, release is done physically onboard the vessel and does not require a password because the equipment is already in a user-access-controlled area. It is not sitting on an open deck where anybody can access it. So there is physical security as well. EXP-01 [Q8/Q10]: Yes. We have already had our cybersecurity arrangements audited by a classification-related third party, and we passed. There was also discussion about Modbus TCP/IP communication because release signals were coming from a third party into our system. The important point was what could actually be changed from outside. You cannot directly change how the pump unit operates or its firefighting capacity. You may be able to affect whether a section is opened or not, so you can do some harm, but not the kind of harm that compromises the complete system. That is how we do it. EXP-01 [Q11]: I think it is quite a good pipeline. We have already done many of these things, so not all of it is new to us, but it brings together many things that need to be considered. Especially the cybersecurity part is a good summary of what is listed in the standards, such as audit logging and monitoring. If you have user roles or device access, and people log in or log out, those events should be recorded in the logs. Password requirements and related security events also need to be handled. EXP-02 [Q1]: Yes, I think it covers them, and perhaps even more than that. It felt like everything you could imagine was included. It could be simplified a bit, but it is good that most of the things were present, including what you showed me in the app. EXP-02 [Q3]: I think this framework is realistic enough, yes. EXP-02 [Q6]: Especially when you showed me how the network works in the schematic diagram in Section 6, I understood much better how it works. EXP-02 [Q8]: To be honest, I did not go into this part in much detail. I only read it quickly, but it makes sense. I think it is okay. EXP-02 [Q13]: There were so many things. At first sight, as I said, you had included almost everything you could imagine. Perhaps going through it and simplifying it a bit would make it more user-friendly. But that is just a minor thing. EXP-02 [Q14]: Cybersecurity will always be a concern for people whenever there is an internet connection to somewhere. The question is how it can be done so that it is as secure as possible. But you have already covered many of these points in your thesis. There is also the business side: how can this be monetised, is it expensive, and who pays for it? What kind of data can we receive? Could data from, for example, pump units be provided or sold to the original equipment manufacturers? EXP-03 [Q1]: Yes, it does. It contains more than a normal system would need. EXP-03 [Q2]: No major ones. There are certainly improvements, as we discussed before the interview, but overall I would give the framework about eight out of ten for EXP-03 [Q2]: It is certainly complex, but I think it is necessary complexity. I would not be too concerned about the complexity itself. EXP-03 [Q2]: Not at the moment. The layers look quite good. I do not have any further comment on that. We have also discussed that this is a general framework and deliberately does not reproduce a confidential company-specific architecture, so for that purpose I would say it is very solid. In an implementation-specific version, some of the naming or allocation of functions and zones

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			EXP-03 [Q3]: Yes. This is an interesting point because the normal industrial practice is that the safest way to protect the system is not to connect it to the internet at all. Realistically, though, with this framework you could potentially convince the customer to move toward a connected system if the benefits are clear. For example, the customer could get real-time monitoring, and maintenance-level access could support diagnostics and troubleshooting without giving direct operational control. That could reduce system downtime, so there is a realistic benefit that could help justify the connection. There is also a wider concern. If the customer allows external connections not only for the fire-suppression system but also for many other critical onboard systems, the ship could end up with many separate external connection points. Even if every connection is protected, having many gateways increases the overall exposure. If one of them is compromised, it could become a serious problem for the vessel. EXP-03 [Q4]: As a stand-alone supplier system, the framework is very good. But in the future, I think ships may combine the externally connected critical systems through one protected ship-level connection point or server. That server could collect the necessary information from systems such as fire suppression, machinery, and energy monitoring, and then external applications or suppliers would obtain selected data through that protected ship-level interface. In my view, that could be stronger than every individual vendor maintaining its own external connection, because the vessel or system integrator can protect and control one EXP-03 [Q8]: Yes, they do. I think the FMVEA-based assessment is very good. You have identified the scenarios and divided them into ratings based on severity and the consequences if an attack or fault succeeds. That structure makes sense. EXP-03 [Q9/Q10]: That comes back to the point we already discussed. You should identify the most common risks and then apply them to the framework to see how the architecture reacts. I think that scenario-based demonstration is not presented strongly enough yet. At the moment, though, the framework looks quite good and appears to protect the system against many common EXP-03 [Q11]: Yes. For now, it would be a good basis for developing a system. In the longer term, as I mentioned, I think critical onboard systems may increasingly connect through a common protected ship-level server, and the fire-suppression system could then exchange selected data through that interface. EXP-03 [Q12]: Network segmentation would be one of the first things to clarify. You should also present specific risk-assessment scenarios and show how the framework responds to them. Customers may not care about every technical detail of how the architecture works. They need to understand what benefit it gives them. For example, you could show a realistic attack or failure scenario and demonstrate how the framework protects the system or reduces downtime. They will also care about the economic side: how much it costs, how quickly it can be implemented, and whether it is worth adding to a system that already works well. EXP-04 [Q10]: Yes. If it is sufficiently protected, I think it could. EXP-04 [Q12]: I think they would receive it quite positively. EXP-04 [Q12]: It can be done, but it needs to be explained clearly, separated into understandable parts, and made simpler for the customer. Not everyone who needs to understand it will have a technical background. EXP-04 [Q12]: The biggest problem is that there are no references yet. It is difficult to sell something that has not been used before. Cybersecurity is very important, so larger companies could definitely be interested, especially if the price and the financial case are EXP-04 [Q13]: The benefits need to be made clearer. It should be explained in which situations this would be the best decision for the customer. EXP-05 [Q1]: It looks concise and good. It contains the basic things we have, including connection to third-party systems. EXP-05 [Q2]: I think it is quite simplified. EXP-05 [Q4]: Yes. Temporary access should also be considered when service personnel go to the customer site. EXP-05 [Q11]: It could be implemented for most vessels, except military vessels. EXP-05 [Q12]: For yachts, privacy may also be a EXP-06 [Q1]: Yes. I think it is pretty well covered. The different layers are laid out nicely, so it is easy to understand how the system is distributed through the framework. I think it contains the required information. EXP-06 [Q2]: No, I do not think so. I think the architecture itself is clear. The complexity is more in the interfaces. EXP-06 [Q12]: The customer approach varies a lot, and it depends on who the point of contact is. In my experience, these cybersecurity and remote-connectivity discussions often need to involve different people than the technical contacts we normally work with. The people responsible for cybersecurity need to be involved so that they can discuss the requirements with people who speak the same technical language. At some point, the people working on the technical side also need to be involved. A concrete demonstration like the one you showed is very useful because then people can see what the concept actually means. If we only show diagrams and layout drawings, the solution is not very tangible. It is harder to understand, and progress can be slow. EXP-06 [Q14]: With my knowledge, I would say that it looks good. I am not the right person to judge in detail whether every part of the cybersecurity architecture is technically correct, so it should also be evaluated by people with deeper expertise in that field. There may be details or weaknesses that still need to be resolved and checked against the relevant requirements, but in the big picture, I think the concept looks good. EXP-07 [Q1]: Yes, I think it is quite good. It contains all the main parts and systems, and I believe it can be adapted quite easily to different implementations of a fire-suppression system.

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			EXP-07 [Q2]: Not much. There is one point that may only be about how the schematic is represented. The connection to the data server and the maintenance gateway could physically be one device while remaining logically separated. That may simply be a matter of presentation and clarity. Otherwise, I do not see unnecessary complexity. I think the framework is detailed enough that implementation work could be started from it. EXP-07 [Q5]: Direct physical access to the safety functions could still compromise them. Apart from that, at least at the conceptual level, it does not appear easy to compromise the essential functions if the proposed separation is maintained. EXP-07 [Q6/Q7]: It can be, but quite often customers require the fire-suppression system to remain fully isolated from the internet. That is an important practical limitation and depends very much on the vessel or site. EXP-07 [Q11]: Yes. I think it would be very useful and could be used as a baseline or starting point for an industrial project. EXP-08 [Q10]: In principle, yes. It is at least a good starting point, but an actual system would still have to be examined in more detail. EXP-08 [Q11]: Yes. Referring to the previous answer, it is at least a good starting point. These points should be considered first, and then the project should examine whether anything more is required for the specific system. EXP-08 [Q13]: I think the key points have been considered. As a concept, the architecture is sound. I do not immediately see a major missing issue. EXP-09 [Q2]: I do not think it looks too complicated for the people who would actually handle the system. They normally already understand this type of system, so it should be relatively easy for them to learn. Tailoring can be useful when possible, but from our side standardisation would also be preferable. EXP-09 [Q6]: I think they can be implemented if the company decides to invest the required time and effort. Related cybersecurity and certification work is already being considered internally, so I do not see why this could not be developed. The detailed technical feasibility should, however, be confirmed by the technical team. EXP-09 [Q7]: The main concern is the possibility of a security breach when access is provided. Normally we do not want to give customers broad access to the system. If access is limited mainly to monitoring, the users are clearly responsible, and passwords and permissions are properly managed, then it should be manageable. It would still create some additional responsibility for the after-sales organisation, and it would probably not be suitable for every vessel or every customer. EXP-09 [Q11]: Yes. It represents the existing system with additional cybersecurity and connectivity functions. If cybersecurity compliance and certification requirements are incorporated, I think an industrial implementation would likely be very close to this framework, even if the final design is not exactly identical. EXP-09 [Q13]: The main improvement would be simplification. The simpler the system and its presentation can be made while still keeping the critical functions, the better. If there are elements that are not necessary, they should be removed. EXP-10 [Q1]: Yes. I think it contains the relevant items very well, at least based on the fire-suppression systems I am familiar with. EXP-10 [Q2]: Yes, the layers make sense. The architecture starts from the field instruments, then moves to the upper levels, and finally includes the external connection possibilities. Although the system itself is complex, the diagram represents it very clearly. EXP-10 [Q11]: Yes. The concepts are very similar to the systems used in practice: there are field instruments, controllers, networks, and the higher-level functions. The framework could therefore be useful for future industrial projects. EXP-10 [Q12]: The concept itself is sufficient, but approvals are an important part of implementation. Customers also need the supplier system to fit into the vessel's complete cybersecurity architecture. The required evidence and documentation are defined by the relevant classification-society rules, so the requirements need to be reviewed class by class. Type approval or an equivalent recognised approval would make deployment easier and could reduce repeated project-specific work. One point I would consider is whether remote access could use a mobile network while the vessel is in port. Ships visit ports frequently, so a separate mobile connection could provide another maintenance-access option without relying on the vessel's own internet connection, provided that such an arrangement is permitted and securely implemented. EXP-10 [Q13]: I cannot identify anything in the proposed concept that clearly needs to be changed. The main practical point I would still investigate is the assumption of internet connectivity and whether alternative, isolated connectivity could be used when appropriate, for example during port visits. EXP-11 [Q1]: Yes. The structure looks very clear to me. It represents the current type of system quite well, but now it is organised into different layers, with the monitoring and maintenance functions added on top. EXP-11 [Q2]: I think the responsibilities are clear. The architecture is very clearly presented, and I do not think it is unnecessarily complex. EXP-11 [Q3]: At least in theory, I think so. The existing functions are already organised in a similar way, and the cyber-resilience and connectivity functions could be added around them. EXP-11 [Q6]: Yes. Most of these measures are realistic, and several are already present in some form. We already use password protection for certain functions, controlled VPN-based access for remote connections, network separation, restricted communication, internal monitoring and logging, version and change management, and protection of controller configurations. Recovery is currently supported through local backups, such as backup storage media. Controlled remote connectivity is also realistic. Some of the exact authentication mechanisms may still need further development, but overall the measures are achievable.

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			EXP-11 [Q7]: I do not think they create major technical difficulties. Cybersecurity will require more procedures and defined ways of working, but technically I do not see it as unnecessary or impractical. EXP-11 [Q7]: Yes, it is realistic. Similar remote-access arrangements have already been implemented in land-based applications using comparable controller technology, so the basic approach is feasible. EXP-11 [Q10]: I am not a specialist in this area, but from my perspective the safeguards look good at the conceptual level. EXP-11 [Q13]: I do not see any clear weakness in the architecture itself. The next useful step would be to build some kind of pilot and use that implementation to identify any remaining gaps that need to be fixed. EXP-11 [Q14]: I cannot add much more detail. Cybersecurity is a difficult topic and people often understand individual parts of it differently. That is why I think it is useful to have a clear overall view like this. The framework gives a good overview of how the different cybersecurity aspects fit together. EXP-12 [Q1]: Yes. I think it describes very well what we have learned and what has been used in recent systems. It contains what I consider relevant at the moment. EXP-12 [Q2]: No. Of course, you can always go further into the details and make the architecture correspond more closely to a specific implementation, but as a generalised framework I think it is very good. EXP-12 [Q6]: In principle, yes, but the controls need to be applied at the right level. For a connected monitoring or cloud service, stronger user management and cybersecurity controls are clearly needed. At the device level, however, implementing local user management on PLC-based equipment may add too much complexity and may not even be practical with the current device capabilities. There is also a trade-off. If local user management requires a connection to supplier servers or to the vessel IT environment, that new connection can itself introduce additional risk. It could also turn each project into a broader IT integration project. For simple equipment-level control of pumps, valves, and local controllers, it may therefore be better to keep the control system simpler and use other protective measures rather than force every whole-vessel IT requirement down to the device level. EXP-12 [Q12]: Remote connectivity changes the approval case the most. Micro-segmentation using managed or programmable network devices may also be needed and can make the system more configurable and secure. The remote-connectivity part is less mature, so it would need to be updated, explained, and assessed with the classification societies, especially where remote actions are possible. The framework gives a good starting point, but those remote functions need explicit assessment.
		Further verification / implementation constraints	EXP-01 [Q6]: Those are more on the software side, for example SCADA and PLC systems. The cybersecurity requirements talk about user profiles and passwords, but for the fire-suppression panel, it should always be available so that you can immediately release or operate the system. You should not need to log in first. That is one of the difficult points when applying the standards. At the moment, we do not really have full role-based access. We mainly have one service access level for doing updates or similar work, and we are even considering whether that could be avoided. Passwords also create additional requirements: password length, failed-login handling, logging of failed attempts, and so on. All of that adds requirements to the software. For normal operation, release is done physically onboard the vessel and does not require a password because the equipment is already in a user-access-controlled area. It is not sitting on an open deck where anybody can access it. So there is physical security as well. EXP-01 [Q8/Q10]: Yes. We have already had our cybersecurity arrangements audited by a classification-related third party, and we passed. There was also discussion about Modbus TCP/IP communication because release signals were coming from a third party into our system. The important point was what could actually be changed from outside. You cannot directly change how the pump unit operates or its firefighting capacity. You may be able to affect whether a section is opened or not, so you can do some harm, but not the kind of harm that compromises the complete system. That is how we do it. EXP-01 [Q11]: I think it is quite a good pipeline. We have already done many of these things, so not all of it is new to us, but it brings together many things that need to be considered. Especially the cybersecurity part is a good summary of what is listed in the standards, such as audit logging and monitoring. If you have user roles or device access, and people log in or log out, those events should be recorded in the logs. Password requirements and related security events also need to be handled. EXP-01 [Q13]: Maybe simplify the zones and layers a little bit. EXP-01 [Q14]: Try to highlight the physical user access as well. The architecture is currently more focused on the software side, but it is quite simple to add the requirement that access points to the system should be located in user-access-controlled areas. There is also the practical issue of physical ports. The standard says that unused open ports should be disabled. In practice, if a panel is installed inside a console or another user-access-controlled location, physically reaching and manipulating that connection may already require a significant amount of effort. So the physical-access EXP-02 [Q2]: That is a good question. With such a short review period, there could surely still be something missing, but it would require a real deep dive into the thesis to notice what is missing or what could be improved. EXP-02 [Q8]: To be honest, I did not go into this part in much detail. I only read it quickly, but it makes sense. I think it is okay. EXP-02 [Q14]: Cybersecurity will always be a concern for people whenever there is an internet connection to somewhere. The question is how it can be done so that it is as secure as possible. But you have already covered many of these points in your thesis. There is also the business side: how can this be monetised, is it expensive, and who pays for it? What kind of data can we receive? Could data from, for example, pump units be provided or sold to the original equipment manufacturers?

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			EXP-03 [Q3]: Yes. This is an interesting point because the normal industrial practice is that the safest way to protect the system is not to connect it to the internet at all. Realistically, though, with this framework you could potentially convince the customer to move toward a connected system if the benefits are clear. For example, the customer could get real-time monitoring, and maintenance-level access could support diagnostics and troubleshooting without giving direct operational control. That could reduce system downtime, so there is a realistic benefit that could help justify the connection. There is also a wider concern. If the customer allows external connections not only for the fire-suppression system but also for many other critical onboard systems, the ship could end up with many separate external connection points. Even if every connection is protected, having many gateways increases the overall exposure. If one of them is compromised, it could become a serious problem for the vessel. EXP-03 [Q8]: Yes, they do. I think the FMVEA-based assessment is very good. You have identified the scenarios and divided them into ratings based on severity and the consequences if an attack or fault succeeds. That structure makes sense. EXP-03 [Q9/Q10]: That comes back to the point we already discussed. You should identify the most common risks and then apply them to the framework to see how the architecture reacts. I think that scenario-based demonstration is not presented strongly enough yet. At the moment, though, the framework looks quite good and appears to protect the system against many common EXP-03 [Q12]: Network segmentation would be one of the first things to clarify. You should also present specific risk-assessment scenarios and show how the framework responds to them. Customers may not care about every technical detail of how the architecture works. They need to understand what benefit it gives them. For example, you could show a realistic attack or failure scenario and demonstrate how the framework protects the system or reduces downtime. They will also care about the economic side: how much it costs, how quickly it can be implemented, and whether it is worth adding to a system that already works well. EXP-04 [Q12]: The biggest problem is that there are no references yet. It is difficult to sell something that has not been used before. Cybersecurity is very important, so larger companies could definitely be interested, especially if the price and the financial case are EXP-05 [Q11]: It could be implemented for most vessels, except military vessels. EXP-05 [Q12]: For yachts, privacy may also be a ----- EXP-06 [Q6]: Authentication and access control: yes. Network separation or segmentation: I am not an expert in that field, but I think yes. Restricted communication belongs to the same area and should be fine. Controlled maintenance access should also be possible. Monitoring and logging may become more challenging once the data moves outside our own system. Up to a certain point we can keep monitoring and logging within the supplier system, but in many cases the customer will be managing the data they receive. Then questions arise about how they log it, how we can use it, and how the monitoring responsibility is divided. Configuration and change management is doable, but it requires a lot of effort. The marine industry is still quite conservative, and these kinds of practices are not equally common across all systems, especially on older vessels. For protection of controller configurations, I am not sure enough to comment in detail. Controlled remote connectivity is always a challenge with moving assets such as vessels, but connectivity onboard ships has taken a significant leap during the last ten years and I think that development will continue. In the EXP-06 [Q12]: The customer approach varies a lot, and it depends on who the point of contact is. In my experience, these cybersecurity and remote-connectivity discussions often need to involve different people than the technical contacts we normally work with. The people responsible for cybersecurity need to be involved so that they can discuss the requirements with people who speak the same technical language. At some point, the people working on the technical side also need to be involved. A concrete demonstration like the one you showed is very useful because then people can see what the concept actually means. If we only show diagrams and layout drawings, the solution is not very tangible, it is harder to understand, and progress can be slow. EXP-06 [Q14]: With my knowledge, I would say that it looks good. I am not the right person to judge in detail whether every part of the cybersecurity architecture is technically correct, so it should also be evaluated by people with deeper expertise in that field. There may be details or weaknesses that still need to be resolved and checked against the relevant requirements, but in the big picture, I think the concept looks good. EXP-07 [Q2]: Not much. There is one point that may only be about how the schematic is represented. The connection to the data server and the maintenance gateway could physically be one device while remaining logically separated. That may simply be a matter of presentation and clarity. Otherwise, I do not see unnecessary complexity. I think the framework is detailed enough that implementation work could be started from it. EXP-07 [Q5]: Yes, that is realistic. Redundancy of this kind is already an established requirement in relevant applications and has been used in systems for decades. EXP-07 [Q6/Q7]: It can be, but quite often customers require the fire-suppression system to remain fully isolated from the internet. That is an important practical limitation and depends very much on the vessel or site. EXP-07 [Q6]: Yes, I think most of these measures can be practically implemented. In some areas, however, we depend heavily on the equipment manufacturers. Certain required features may be limited by the capabilities of the devices being used. For example, with configuration and change management, we may not be able to implement device-level tracking of configuration changes if the equipment does not support it. In that case, we have to manage the change-control process on our own side rather than relying on EXP-07 [Q13]: The architecture itself is quite good. The main question is whether, in a specific implementation, we can actually separate all of the zones and layers as cleanly as the framework shows. For example, in our type of system, access to the Supervisory Monitoring Layer may also provide a path toward the Communication Network Layer. It is not necessarily a complete compromise, but the layers are not always independent in practice. That implementation-level overlap needs to be considered.

RQ	Theme	Code	Quotation
			EXP-08 [Q8]: Yes, they look realistic. The assessment may not be complete, because it may not be possible to imagine every type of threat in advance. EXP-08 [Q10]: In principle, yes. It is at least a good starting point, but an actual system would still have to be examined in more detail. EXP-08 [Q11]: Yes. Referring to the previous answer, it is at least a good starting point. These points should be considered first, and then the project should examine whether anything more is required for the specific system. EXP-08 [Q12]: I cannot immediately say that something specific must be changed. The framework would need to be tested against an actual project to see whether it is sufficient, while also considering all applicable standards and rules. That is the realistic way to EXP-09 [Q2]: I do not think it looks too complicated for the people who would actually handle the system. They normally already understand this type of system, so it should be relatively easy for them to learn. Tailoring can be useful when possible, but from our side standardisation would also be preferable. EXP-09 [Q6]: I think they can be implemented if the company decides to invest the required time and effort. Related cybersecurity and certification work is already being considered internally, so I do not see why this could not be developed. The detailed technical feasibility should, however, be confirmed by the technical team. EXP-09 [Q7]: The main concern is the possibility of a security breach when access is provided. Normally we do not want to give customers broad access to the system. If access is limited mainly to monitoring, the users are clearly responsible, and passwords and permissions are properly managed, then it should be manageable. It would still create some additional responsibility for the after-sales organisation, and it would probably not be suitable for every vessel or every customer. EXP-09 [Q11]: Yes. It represents the existing system with additional cybersecurity and connectivity functions. If cybersecurity compliance and certification requirements are incorporated, I think an industrial implementation would likely be very close to this framework, even if the final design is not exactly identical. EXP-09 [Q14]: The architecture should also be checked carefully against the cybersecurity requirements and guidance of the relevant classification society. I expect the underlying requirements to be similar to the standards already used in the framework, but this should still be verified explicitly. Otherwise, the interface and overall concept look good and relatively simple. EXP-10 [Q12]: The concept itself is sufficient, but approvals are an important part of implementation. Customers also need the supplier system to fit into the vessel's complete cybersecurity architecture. The required evidence and documentation are defined by the relevant classification-society rules, so the requirements need to be reviewed class by class. Type approval or an equivalent recognised approval would make deployment easier and could reduce repeated project-specific work. One point I would consider is whether remote access could use a mobile network while the vessel is in port. Ships visit ports frequently, so a separate mobile connection could provide another maintenance-access option without relying on the vessel's own internet connection, provided that such an arrangement is permitted and securely implemented. EXP-10 [Q13]: I cannot identify anything in the proposed concept that clearly needs to be changed. The main practical point I would still investigate is the assumption of internet connectivity and whether alternative, isolated connectivity could be used when appropriate, for example during port visits. EXP-11 [Q7]: I do not think they create major technical difficulties. Cybersecurity will require more procedures and defined ways of working, but technically I do not see it as unnecessary or impractical. EXP-11 [Q7]: Yes, it is realistic. Similar remote-access arrangements have already been implemented in land-based applications using comparable controller technology, so the basic approach is feasible. EXP-11 [Q10]: I am not a specialist in this area, but from my perspective the safeguards look good at the conceptual level. EXP-11 [Q11]: Yes. I think it can be used as a high-level guideline. EXP-11 [Q12]: I would still want the concept to be reviewed with a relevant system or cybersecurity architect. A short technical review with that expertise would help confirm the details before moving EXP-11 [Q13]: I do not see any clear weakness in the architecture itself. The next useful step would be to build some kind of pilot and use that implementation to identify any remaining gaps that need to be fixed. EXP-12 [Q6]: In principle, yes, but the controls need to be applied at the right level. For a connected monitoring or cloud service, stronger user management and cybersecurity controls are clearly needed. At the device level, however, implementing local user management on PLC-based equipment may add too much complexity and may not even be practical with the current device capabilities. There is also a trade-off. If local user management requires a connection to supplier servers or to the vessel IT environment, that new connection can itself introduce additional risk. It could also turn each project into a broader IT integration project. For simple equipment-level control of pumps, valves, and local controllers, it may therefore be better to keep the control system simpler and use other protective measures rather than force every whole-vessel IT requirement down to the device level. EXP-12 [Q11]: Yes, to some extent. The important thing is not to add unnecessary complexity. A high-volume system supplier needs to meet the requirements without over-engineering every project. The framework appears to contain the necessary elements, but I would still question each element and ask whether it is genuinely needed in the specific implementation.

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			EXP-12 [Q12]: Customers will want confidence that the system can obtain the required IACS UR E27 compliance evidence or certification and that the documentation is complete. For a fire-suppression system, many customers are not yet asking for extensive IoT or remote-monitoring functionality; that is still relatively new. The demand is likely to increase as customers become more interested in mobile or fleet-level monitoring. EXP-12 [Q12]: First, I would define the micro-segmentation more clearly. Second, I would link the architecture directly to the applicable requirements. If we have an IT or control-system topology and a cybersecurity design, I want to be able to point from each relevant IEC 62443 requirement to the architectural feature that addresses it. The same traceability is important for IACS UR E27 and classification-society approval. If the architecture is taken into use, the next step is to build the approval documentation from those EXP-12 [Q12]: Remote connectivity changes the approval case the most. Micro-segmentation using managed or programmable network devices may also be needed and can make the system more configurable and secure. The remote-connectivity part is less mature, so it would need to be updated, explained, and assessed with the classification societies, especially where remote actions are possible. The framework gives a good starting point, but those remote functions need explicit assessment.
	Refinement / simplification recommended		EXP-01 [Q13]: Maybe simplify the zones and layers a little bit. EXP-01 [Q14]: Try to highlight the physical user access as well. The architecture is currently more focused on the software side, but it is quite simple to add the requirement that access points to the system should be located in user-access-controlled areas. There is also the practical issue of physical ports. The standard says that unused open ports should be disabled. In practice, if a panel is installed inside a console or another user-access-controlled location, physically reaching and manipulating that connection may already require a significant amount of effort. So the physical-access EXP-02 [Q1]: Yes, I think it covers them, and perhaps even more than that. It felt like everything you could imagine was included. It could be simplified a bit, but it is good that most of the things were present, including what you showed me in the app. EXP-02 [Q2]: That is a good question. With such a short review period, there could surely still be something missing, but it would require a real deep dive into the thesis to notice what is missing or what could be improved. EXP-02 [Q13]: There were so many things. At first sight, as I said, you had included almost everything you could imagine. Perhaps going through it and simplifying it a bit would make it more user-friendly. But that is just a minor thing. EXP-03 [Q1]: Yes, it does. It contains more than a normal system would need. EXP-03 [Q2]: No major ones. There are certainly improvements, as we discussed before the interview, but overall I would give the framework about eight out of ten for EXP-03 [Q2]: It is certainly complex, but I think it is necessary complexity. I would not be too concerned about the complexity itself. EXP-03 [Q2]: Not at the moment. The layers look quite good. I do not have any further comment on that. We have also discussed that this is a general framework and deliberately does not reproduce a confidential company-specific architecture, so for that purpose I would say it is very solid. In an implementation-specific version, some of the naming or allocation of functions and zones EXP-03 [Q9/Q10]: That comes back to the point we already discussed. You should identify the most common risks and then apply them to the framework to see how the architecture reacts. I think that scenario-based demonstration is not presented strongly enough yet. At the moment, though, the framework looks quite good and appears to protect the system against many common EXP-04 [Q12]: It can be done, but it needs to be explained clearly, separated into understandable parts, and made simpler for the customer. Not everyone who needs to understand it will have a technical background. EXP-05 [Q2]: I think it is quite simplified. EXP-05 [Q4]: Yes. Temporary access should also be considered when service personnel go to the customer site. EXP-06 [Q2]: No, I do not think so. I think the architecture itself is clear. The complexity is more in the interfaces. EXP-07 [Q2]: I could combine the data-server and maintenance-gateway functions physically, as I mentioned, while keeping them logically separated. In this particular type of fire-protection system, the Communication Network Layer, Safety Function Layer, and Supervisory Monitoring Layer also have quite a lot of overlap and may be partly combined in practice. But as a conceptual way of presenting the architecture, I think the current separation is better for understanding it. EXP-07 [Q3]: Yes, I think it is mostly realistic. One difficulty is that the Safety Function Zone and the physical fire-suppression functions cannot always be fully separated in a real implementation. In this particular case, if you gain access to some safety-control functions, that may also provide access toward the fire-suppression functions. So that boundary can be difficult to realise exactly as shown, but otherwise the separation is realistic. EXP-08 [Q2]: I do not think so. There may even be more in the framework than what would actually be needed in a specific implementation. EXP-09 [Q12]: For non-technical and commercial people, I would want a very simple diagram showing what the current system looks like and what is being added in the future. It should be clear that this is not a completely new fire-suppression system; it is additional cybersecurity and connectivity around the existing system. The simpler that comparison is, the easier it will be to explain internally and to customers. When presenting to a customer, the relevant electrical or engineering people can then review the technical details. A cybersecurity and connected-service offering could provide commercial value because not every competitor offers the same capability or certification. It could be sold as an additional option for new projects and potentially as an upgrade for the existing installed base. Even if the additional hardware cost per vessel is relatively modest, applying the upgrade across a large

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			EXP-09 [Q13]: The main improvement would be simplification. The simpler the system and its presentation can be made while still keeping the critical functions, the better. If there are elements that are not necessary, they should be removed. EXP-09 [Q14]: The architecture should also be checked carefully against the cybersecurity requirements and guidance of the relevant classification society. I expect the underlying requirements to be similar to the standards already used in the framework, but this should still be verified explicitly. Otherwise, the interface and overall concept look good and relatively simple. EXP-10 [Q2]: Yes, the layers make sense. The architecture starts from the field instruments, then moves to the upper levels, and finally includes the external connection possibilities. Although the system itself is complex, the diagram represents it very clearly. EXP-11 [Q2]: I think the responsibilities are clear. The architecture is very clearly presented, and I do not think it is unnecessarily complicated. EXP-11 [Q4]: Yes. I think the communication paths are clear. The way they are represented, including the different connection types, makes the communication between the areas easy to understand. EXP-12 [Q2]: Not in the main schematic. I do not think that view is too complex. In some of the earlier layer descriptions, however, I did not fully see the need to separate the system into so many levels. Extra layers add complexity if there is no clear reason for them, so I would question whether every separate layer is really needed. EXP-12 [Q3]: Fire suppression is part of the safety function in my opinion. The system has a fairly specific purpose: fire suppression. I would therefore consider combining the separate safety-function and fire-suppression areas into one fire-suppression functionality zone. If they remain separate, the contents of each zone need to be described more clearly. The supervisory area could be intended for authorised supplier personnel, while maintenance or engineering access from a third party should be lower-trust and more EXP-12 [Q3]: Yes. The main schematic describes it well. I would still simplify some of the other layer or zone definitions where the separation does not provide a clear benefit. EXP-12 [Q6]: In principle, yes, but the controls need to be applied at the right level. For a connected monitoring or cloud service, stronger user management and cybersecurity controls are clearly needed. At the device level, however, implementing local user management on PLC-based equipment may add too much complexity and may not even be practical with the current device capabilities. There is also a trade-off. If local user management requires a connection to supplier servers or to the vessel IT environment, that new connection can itself introduce additional risk. It could also turn each project into a broader IT integration project. For simple equipment-level control of pumps, valves, and local controllers, it may therefore be better to keep the control system simpler and use other protective measures rather than force every whole-vessel IT requirement down to the device level. EXP-12 [Q11]: Yes, to some extent. The important thing is not to add unnecessary complexity. A high-volume system supplier needs to meet the requirements without over-engineering every project. The framework appears to contain the necessary elements, but I would still question each element and ask whether it is genuinely needed in the specific implementation.